

Ideas and Firm Dynamics

When It Takes Two to Tango

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Abstract

Teams of inventors now produce over 80% of U.S. patents, up from just 45% in 1976. Over the same period, inventor employment has become increasingly concentrated in large firms, rising six times faster than overall employment concentration. To explain these trends, we develop a model of endogenous team formation within firms and embed it in a general equilibrium model of the inventor labor market. Teams increase firms' returns to scale by allowing inventors to select from a larger pool of potential collaborators, increasing the likelihood of productive matches. Rising knowledge specialization over time increases the value of these matches and, in turn, the gains from team production. We estimate the model using the patenting output and inventor employment of the universe of U.S. patenting firms and find that rising team productivity can account for the observed increase in inventor employment concentration. We then study the welfare implications of this shift. Because knowledge spillovers across firms are not internalized, the competitive allocation of inventors is inefficient. The rise of team production amplifies this misallocation by a factor of seven, reallocating inventors away from small firms that generate disproportionately large knowledge spillovers.

Keywords: innovation, R&D teams, inventor allocation, firm dynamics, economic growth

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1 Introduction

The development of new ideas is a key driver of economic growth. Ideas are produced either by individual inventors or by collaborations among teams of inventors. Teams have become increasingly important in idea production in recent decades, both in patenting and in scientific research ([Wuchty, Jones, and Uzzi 2007](#)). Existing models of inventor team formation study important dimensions of collaboration—such as specialization, sorting, human capital accumulation, and declining communication costs—but abstract from firm boundaries.

The central premise of this paper is that firm boundaries impose meaningful constraints on inventor collaboration. Changes to the idea production function that favor teams such as increasing specialization ([Jones 2009](#)) or falling communication costs ([Pearce 2023](#)) may therefore reshape the allocation of inventors to firms in order to exploit collaborations among inventors. When teams are the primary unit of knowledge production and team formation is constrained by the boundary of the firm, modeling the two jointly is essential for studying trends in the allocation of inventors to firms and its implications for the rate of innovative output. Inventors are also increasingly concentrated at large firms, raising concerns about the implications for growth ([Akcigit and Goldschlag 2023b](#); [Fernandez-Villaverde, Yu, and Zanetti 2026](#)).

Our main contribution is to propose a novel and tractable framework for modeling team-based invention within firms in which additional inventor colleagues raise expected inventor productivity only in firms that choose to adopt a team production model. We then measure the impact of team production on innovative output (patents) using a panel of the universe of patenting firms in the U.S. Consistent with the theory, we show that team-based firms have less-decreasing returns to scale in inventor labor and the impact of choosing a team production model on returns to scale has increased over time, likely reflecting greater specialization of knowledge ([Freund 2024](#)). In our model, inventors become more concentrated at large firms in response to this shock. Finally, we explore whether this reallocation of inventors is efficient in the presence

of knowledge spillovers. A planner would also choose to increase concentration in response to the measured changes in the idea production function favoring teams, but not nearly as much as happens in the competitive equilibrium. The changes in the idea production function favoring teams that we measure in the data increase misallocation (the gap between innovative output in the socially optimal allocation and the competition equilibrium) by a factor of seven.

Our analysis precedes in several steps. First, using patent data, we document several stylized facts about inventor teams. First, patents developed by teams now account for 80% of all private patents in the U.S., compared with 45% in 1980. Second, around 90% of team collaborations in patenting occur among inventors within the same firm. This underscores the importance of firm boundaries in shaping collaborative idea production. We further find that patenting output per worker, measured by citation-weighted patents, increases with the number of potential collaborators and the intensity of team patenting within the firm. Finally, we document that inventor employment concentration in large firms has risen dramatically over time, and that this rise is more pronounced than the increase in overall employment concentration ([Autor et al. 2019](#)) over the same period by a factor of six.

Motivated by these facts, we build a general equilibrium model of the inventor labor market with heterogeneous firms in the spirit of [Hopenhayn \(1992\)](#) and [Melitz \(2003\)](#), augmented with two new ingredients. First, we explicitly model the process of inventor team formation within firms. Each inventor draws a bilateral productivity for collaborating with every other inventor in the firm and can alternatively work alone with some fixed productivity. The firm optimally assigns inventors across solo and team production to maximize output. As a result, realized inventor productivity is an increasing but concave function of the number of inventor coworkers in the firm, as inventors are increasingly likely to find a productive match with each additional coworker.¹ Second, we incorporate knowledge spillovers across firms that depend on both firm productivity

¹We support these modeling assumptions and the implied relationship between inventor productivity and firm size with empirical evidence from the U.S. patent data.

and firm employment size, following [Chikis, Kleinman, and Prato \(2025\)](#). The presence of spillovers implies that the equilibrium firm size distribution need not be efficient.

Next, we use a combination of direct and indirect inference to estimate how key model parameters, particularly those governing the idea production function for solo inventors and teams, have evolved over time. We then use our model to quantitatively evaluate how these changes affect both the competitive equilibrium allocation and the socially efficient allocation of inventors across firms and assess welfare implications.

We estimate the model for two periods: 1976-1999 and 2000-2018. The estimation combines direct production function estimates of the model's idea production function and knowledge spillover function with indirect inference to match the evolution of team patenting, inventor allocation across firms, and firm dynamics. We find that the average productivity of both solo inventors and inventor teams has risen over time, while the returns to scale benefit of teams, capturing knowledge specialization, has increased. Our model predicts that these changes in the idea production function generate a substantial increase in employment concentration and average firm size, consistent with the data. At the same time, the model predicts higher innovative output but an endogenous decline in knowledge spillovers, consistent with the declining knowledge diffusion hypothesis of [Akcigit and Ates \(2023\)](#).

Finally, we assess the welfare implications of the changes in idea production associated with the rise of patenting teams. We find that inventor misallocation becomes more pronounced as team production expands in the presence of knowledge spillovers. The underlying mechanism is that the measured shifts in the idea production function favor large, highly productive firms, which increases the wage and leads to excessive firm exit and a contraction of smaller firms who generate a disproportionately large amount of knowledge spillovers compared to their innovative output ([Chikis, Kleinman, and Prato 2025](#)). These results have implications for the design of R&D policy. In particular, the welfare gains from subsidizing small firms and firm entry are larger in the current environment than in the past.

Related Literature. A growing body of literature examines the match between individual inventors and firms (Akcigit and Goldschlag 2023a; Akcigit and Goldschlag 2023c; Babalievsky 2023; Manera 2022; Liu 2023; Guccione and Roldan-Blanco 2026). A separate strand of literature models firms as collections of coworkers or co-inventors, without incorporating additional dimensions of firm heterogeneity (Herkenhoff et al. 2024; Pearce 2023; Freund 2024). This paper bridges the gap between these two approaches by exploring how inventor team formation varies with firm-level characteristics (in particular idiosyncratic firm productivity, which is intimately connected in most models, including ours, to employment size) and relates to firm dynamics and inventor (mis)allocation across firms.² This connection is important for understanding how changes in the idea production function might have played a role in the recent rise in employment concentration of inventors.

Our model of teamwork within firms contributes to our understanding of the boundary of the firm and optimal scale (Lucas 1978; Chandler 1977). Recent empirical work finds evidence of differences in returns to scale at the firm level within industries (Chan et al. 2025) and evidence that the supply of managers (Engbom et al. 2025), standardization (Argente et al. 2025), and economic development (Bassi et al. 2023; Chen 2023) all influence optimal firm scale. We contribute an additional microfoundation for changing optimal firm scale: idea production in teams, and validate the existence of this explanation using firm level data on inputs (inventors) and output (patents). Our mechanism may extend beyond patenting to other settings in which workers increasingly specialize along the development path. However, patent data provide a uniquely rich environment to observe team collaborations within firms, which is more challenging to measure in other industries.

Our findings also contribute to understanding trends in productivity differences between large and small firms, particularly in the context of research productivity (Andrews, Criscuolo, and Gal 2016; Bessen et al. 2020; Pugsley, Sedlacek, and Sterk 2021;

²Jarosch, Oberfield, and Rossi-Hansberg (2021) and Boerma, Tsyvinski, and Zimin (2023) study models of sorting into teams within firms, but not specifically for inventors.

Cavenaile, Celik, and Tian 2025; Olmstead-Rumsey 2025; Akcigit and Ates 2023), and the recent trends of declining business dynamism and missing high-growth young firms (Decker et al. 2014; Decker et al. 2016; Sterk, Sedláček, and Pugsley 2021; Akcigit and Ates 2023).

This paper is also related to the literature on how interactions between individuals shape economic growth (Jones 2009; Lucas and Moll 2014; Akcigit et al. 2018; Prato 2022). Relative to these papers, we abstract from the dynamics of individual-level knowledge accumulation and instead focus on the organizational features that determine how knowledge is produced within firms (Garicano and Rossi-Hansberg 2015). Our model delivers the novel prediction that R&D workers increasingly concentrate in large firms, which highlights the importance of understanding how human capital accumulation for R&D workers operates. If exposure to a diverse set of employers over an inventor’s career is a key source of knowledge accumulation, rising concentration in larger firms may pose challenges. On the other hand, the shift from solo invention toward collaboration may accelerate on-the-job human capital accumulation through peer learning within firms (Akcigit et al. 2018). We leave the study of these dynamics for future work.

Finally, several empirical studies highlight the role of inventor teams within firms. Jaravel, Petkova, and Bell (2018) find that the death of a co-inventor negatively impacts subsequent inventor productivity. Bhaskarabhatla et al. (2021) show that a significant share of firm-level heterogeneity in research productivity is driven by differences in inventor quality. Kline et al. (2019) find that firms share around 30% of patenting rents with incumbent high-paid employees. We abstract from these issues and instead focus our attention on how the number of coworkers within a firm affects inventor productivity. We document new facts on patenting output and team size as functions of firm size, and show that organizing inventors into teams raises returns to scale. Our findings also indicate that this effect strengthens over time and carries important welfare implications because of its interaction with the (mis)allocation of workers across firms.

2 Empirical Motivation

2.1 Data Description

Our primary data sources are the United States Patent and Trademark Office (USPTO) PatentsView, Revelio Labs LinkedIn data, and S&P’s Compustat. The USPTO PatentsView database tracks the universe of patents granted by the U.S. Patent and Trademark Office (USPTO) from 1976 onward. It provides detailed patent-level information, including application and grant dates, technology class classifications, inventor identifiers, citations, and the names and addresses of patent assignees (firms).

The Revelio Labs LinkedIn data offer worker-level employment histories based on LinkedIn profiles, which is linked to the USPTO inventor data. This allows us to match inventors to firms in years that they do not patent and hence do not appear in the USPTO data.³

S&P Compustat provides detailed firm-level data for publicly listed firms, maintained by S&P Global. It includes information such as firm size, industry scope, balance sheets, income and cash flow statements, and stock prices.

We collect utility patents filed between 1976 and 2018, granted by the USPTO. To identify the patents assigned to U.S. public firms and track them by different firm characteristics, we link the USPTO assignee identifier to the Compustat GVKEYs (the firm identifiers in Compustat) using the bridge developed by [Ding, Jo, and Kim \(2022\)](#) and [Braguinsky et al. \(2023\)](#).⁴ When a patent is assigned to multiple entities, we only keep the first assignee following the sequence order recorded by USPTO. To measure the quality of patents, we calculate five-year forward citations with vintage fixed effects removed, and obtain the real patent value measures from [Kogan et al. \(2017\)](#) (KPSS values, hereafter) for patents filed by publicly held firms.

³Eventually, we aim to incorporate administrative data from the U.S. Census Bureau to extend our analysis, which is currently in progress.

⁴This bridge is constructed using the standard name and address matching process, complemented by an internet search-aided algorithm following [Autor et al. \(2020\)](#) as well as additional manual matching. The detailed methodology is described in [Ding, Jo, and Kim \(2022\)](#) and [Braguinsky et al. \(2023\)](#).

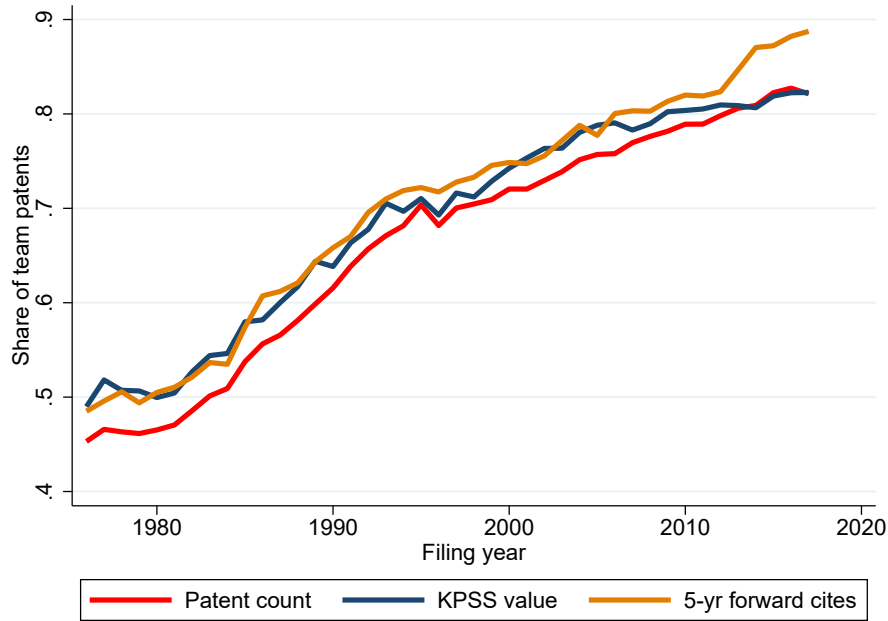


Figure 1: Team Patents Shares

Note: The figure depicts the share of team patents at Compustat firms from 1976 to 2018. A team patent is defined as one assigned by multiple inventors. The number of patents is counted unweighted (red line), weighted by KPSS values from Kogan et al. (2017) (navy line), and weighted by five-year forward citations with vintage fixed effects removed (orange line).

2.2 Empirical Findings

Next, we summarize the core set of empirical findings that emerge from our dataset.

Fact 1: Team patents have increased from 50% (1976) to 80% (2018) of patents

First, team-based patenting has become increasingly prevalent over the past five decades. We define a *team patent* as a patent with multiple inventors, and refer to the set of inventors listed on the patent as a *team*.

In our data, the share of team patents has risen from 45% in 1976 to above 80% in 2018. This upward trend holds either measured in patent counts, market-based patent value (KPSS values), or five-year forward citations. These patterns are illustrated in Figure 1.⁵

⁵In this figure, we restrict our sample to Compustat firms with patent filings given the availability of

Fact 2: 78-90% of team patents are by inventors at the same firm

Second, we find that the vast majority of team collaboration in innovation occurs within firm boundaries rather than across firms. This highlights the role of firms as the critical platform for team production.

In the USPTO data, we identify an inventor's affiliation in a given year as the assignee (firm) with which the inventor files the largest number of patents in that year. In the event of a tie, we assign the affiliation to the firm associated with the patent having the latest filing date. Using this definition, 89.5% of team patents in our sample are produced by inventors affiliated with the same firm. This indicates that most team-based innovation reflects within-firm collaboration.

The Revelio Labs data helps us overcome the challenge of separating employment and credit on a patent which is not possible in the USPTO data. In Revelio, we compute the probability that a given pair of inventors within a team is employed by the same firm within a five-year window centered on the patent filing date. We estimate this probability to be 78.8% on average in our sample. Additional details are provided in Appendix C.1, with Figure C6 showing that the probability of working together in any single month before the filing date peaks around 10 months before the filing date at about 67%.

Fact 3: Inventor employment concentration has increased much more than overall employment concentration

A growing literature analyzes the causes and consequences of rising employment concentration in the United States (Autor et al. 2017; Rossi-Hansberg, Sarte, and Trachter 2020). Akcigit and Ates (2023) were the first to point out the increase in employment concentration specifically among inventors and Akcigit and Goldschlag (2023c) provide additional supporting evidence. To our knowledge, we are the first to provide a direct

KPSS values and to maintain consistency of the sample across series. The results are robust when using the full set of USPTO assignees.

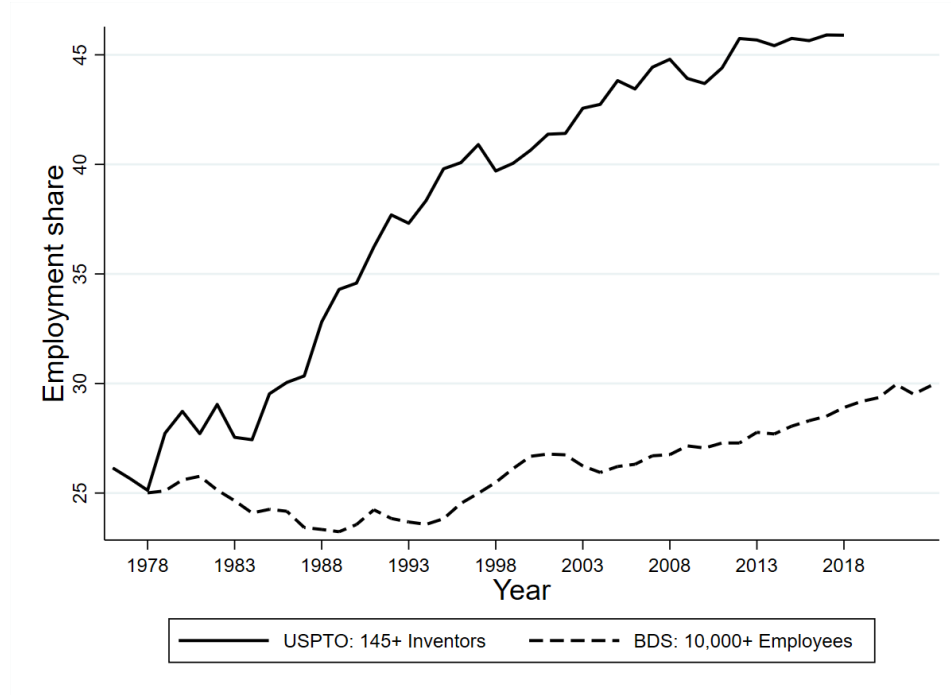


Figure 2: Employment Concentration: Inventors and Overall Employment

Note: The figure plots the employment share of inventors at the largest firms (ranked by the number of inventors), indexed to match the employment share of overall employment at the largest firms (ranked by total employment) in 1978. Inventor employment is measured using patent data in USPTO to count inventors at the firm–year level, while overall employment shares are constructed from the Census Bureau’s Business Dynamics Statistics (BDS).

comparison between overall employment concentration and concentration in inventor employment.

To do this, we use the USPTO data to match inventors to firms (assignees in the patent data).⁶ A standard measure of overall employment concentration is the share of total employment accounted for by firms with more than 10,000 employees reported by the Census Business Dynamics Statistics (Autor et al. 2017). We set a similar threshold for inventor employment by matching the share of inventor employment at large innovative firms in 1978 to the share of overall employment at firms with more than 10,000 employees in 1978. This procedure gives us a threshold of 145 inventors for

⁶This is the same approach taken by Akcigit and Goldschlag (2023c) and means that our measure of inventors includes only those who patent in a given year. The Revelio Labs data provides a broader coverage of inventors, including those who do not patent, and we use it as a supplementary source. However, this data does not span a sufficiently long time horizon to study long-run trends in concentration.

defining large innovative firms.

Holding fixed this threshold over time, the inventor employment share of large innovative firms has risen about six times more than the corresponding overall employment share of large firms. Figure 2 presents this pattern. Our model in the following section explores how this divergence can arise from changes in optimal innovative firm size driven by changes in the idea production function that favor knowledge production in teams.

3 Theoretical Framework

To explain the trends documented in Section 2 and assess their implications for the allocation of inventors across firms, we develop a theoretical framework to study the problem of heterogeneous innovative firms hiring inventors in general equilibrium.

The model has the following three key ingredients: first, we introduce firm heterogeneity in productivity to generate dispersion in firm size as is standard in the firm dynamics literature (Hopenhayn 1992; Melitz 2003); second, we provide a novel microfoundation for the firm’s internal allocation of inventors to either solo or team-based idea production and endogenize the economy’s fraction of team patenting; and third, we allow for research productivity spillovers across firms that depend both on a firm’s intrinsic productivity and its employment size following Chikis, Kleinman, and Prato (2025).

We begin by specifying inventor-level productivity, which depends on whether the inventor works alone or in a team. We then describe the firm’s problem of hiring inventors and deciding whether to organize inventors into solo or team production. Finally, we define and characterize the equilibrium. Appendix A.1 provides additional empirical evidence that supports the key modeling assumptions and results.

3.1 Inventor Productivity

In the initial “brainstorming” phase that determines inventor productivity, an inventor can either work alone (“solo”) or collaborate with another inventor within the same firm (“team”). Inventor i has the following productivity:

$$\begin{cases} z_{ii} = z & \text{if working alone} \\ z_{ij} \sim \text{Pareto}(x, \frac{1}{\gamma}) & \text{if working with another inventor } j. \end{cases} \quad (3.1)$$

When working alone, inventor productivity is z .⁷ Under team production, inventor i can potentially collaborate with any other inventor j within the firm. The productivity of each of these potential collaborations z_{ij} is drawn from a Pareto distribution with scale and shape parameters $(x, \frac{1}{\gamma})$.

A few remarks are in order. First, inventors find potential team members only within the boundary of the firm, consistent with Fact 2. Second, while we do not model team communication costs explicitly, we maintain the assumption throughout that $x < z$, that is, the least productive team collaboration is less productive than working alone, suggesting some costs to working in ill-matched teams.⁸ Third, for simplicity, we assume that each potential team consists of only two inventors. Appendix Figure A1 shows that even as firm size grows large, the mean team size within the firm is stable at around three inventors. Appendix A.2 sketches how allowing for productivity draws for teams of three would strengthen our mechanism because it provides inventors in large firms with more draws over which the firm can take the maximum to determine realized productivity.

Finally, the pairwise productivity draws reflect potential gains from collaboration: inventors may combine their expertise in highly productive ways, for instance through

⁷A simple extension, though one we do not pursue here, would be to allow stochastic productivity for solo inventors with mean z . The firm’s decision of whether to adopt team production would be unchanged.

⁸We also abstract from serial collaboration or team-specific human capital accumulation since our model is static, but this is also consistent with evidence in Appendix A.1 that 80% of teams only patent together once.

complementarities in skills or knowledge. The stochastic structure of pairwise productivity implies heterogeneity in match quality within the firm: not all inventors pair equally well, and not all potential collaborations are equally productive. Appendix A.3 contains a further microfoundation for this setup based on Freund (2024); developing an idea requires many tasks or fields of knowledge over which each inventor has a vector of skills. A parameter χ governs within-worker skill dispersion across tasks, with higher χ capturing greater specialization (Jones 2009; Freund 2024). The quality z_{ij} of the idea a randomly drawn pair of inventors can develop is increasing in the distance between i and j 's skill vectors. Greater specialization χ amplifies the effect of skill distance on output by amplifying the efficiency gains from specialization: the distribution of teams' idea quality has a fatter tail when specialization increases (Figure A4). Increased γ in our model can be thought of as greater specialization of inventors in particular tasks or fields of knowledge needed to develop an idea.

Suppose the firm's objective is to maximize the expected productivity of its inventors (this corresponds to profit maximization in the following section). Since workers are ex ante homogeneous, expected productivity is the same for all inventors in a firm of size n . The firm decides, before the realization of pairwise productivities, whether to adopt solo or team idea production. If it adopts solo idea production, inventors all work alone with productivity z . If it adopts team idea production, the firm can observe all realizations of pairwise productivity among its workers and allocate each worker to their most productive collaboration. Collaborations are not mutually exclusive: we envision teamwork as serving a brainstorming role that affects inventor productivity in the production of ideas in the second stage, so if the same inventor is the best match for multiple coworkers in the brainstorming phase they can all be matched with that inventor. For now we describe expected inventor productivity under team production, which is the relevant object for the firm.

Let z_i^{team} denote the maximal team productivity available to inventor i within the firm. In a firm with a discrete number n inventors, the realized team productivity for

inventor i would be

$$z_i^{team} = \max\{z_{i1}, z_{i2}, \dots, z_{in}\}. \quad (3.2)$$

Given the Pareto distribution of pairwise productivity, the expected team productivity of an inventor in a firm of size n is

$$\mathbb{E}(z_i^{team}) = x\Gamma(1 - \gamma) n^\gamma. \quad (3.3)$$

See Appendix A.4.1 for the derivation. Note that this property also holds with a continuum of workers, as we will have in our general equilibrium model, by interpreting n as the mass of inventors employed by the firm and letting bilateral productivities be drawn from an inhomogeneous Poisson point process. See Appendix A.4.2 for details.

Equation (3.3) implies that expected team productivity is increasing in firm size n , the scale parameter x , and in the thickness of the right tail of the Pareto distribution of pairwise productivity draws γ which reflects the degree of knowledge specialization of inventors.

At the worker level, because of the assumption that $z > x$, team productivity exceeds solo productivity when the worker belongs to a sufficiently large firm. In particular, there exists a firm-size threshold such that inventors in firms with size above the cutoff achieve higher expected productivity in teams than under solo production. Figure 3 depicts this relationship and the cutoff. Consistent with this prediction, Appendix Table A1 documents a positive, economically and statistically significant relationship between innovative firm size (measured by the number of inventors) and the share of team patents in the firm's total patent count.

3.2 Innovative Output at the Firm-Level

In the second stage, firms transform inventor efficiency units—determined during the brainstorming stage—into innovative output using a decreasing-returns production technology (Lucas 1978). A firm's total innovative output is given by:

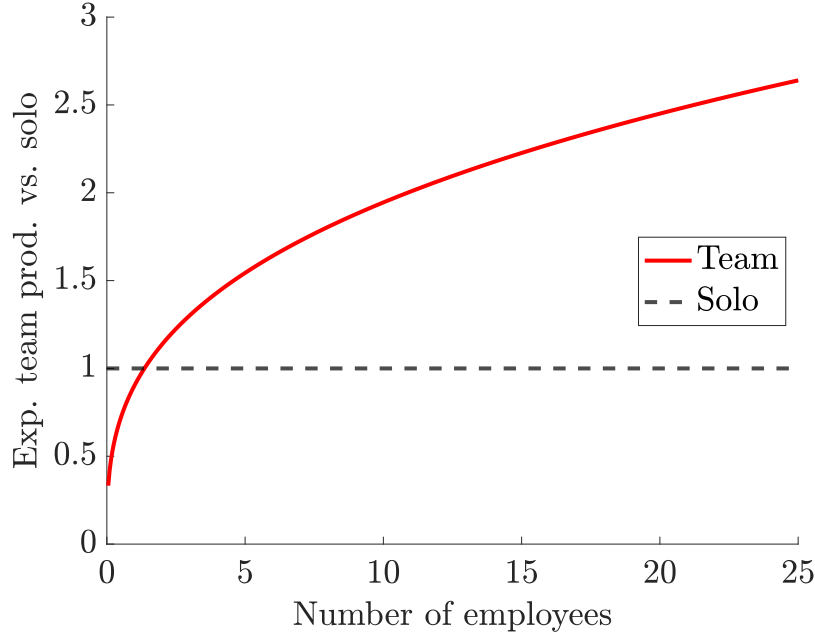


Figure 3: Team vs Solo Productivity

Note: The figure compares worker-level productivity under team and solo production as a function of firm size n . The red solid line indicates team production, while the black dashed line shows solo production.

$$y_f = \begin{cases} \zeta \varepsilon_f (z n_f)^\alpha, & \text{if solo,} \\ \zeta \varepsilon_f (x \Gamma(1 - \gamma) n_f^\gamma n_f)^\alpha, & \text{if team.} \end{cases} \quad (3.4)$$

Firms are heterogeneous in productivity, denoted by ε_f , which is drawn from a Pareto distribution, $\varepsilon_f \sim \text{Pareto}(x_{\min}^f, \frac{1}{\eta})$, with cumulative distribution function G .

The key prediction of the microfoundation is that team-based firms have less-decreasing returns to scale in inventor labor, coming from an agglomeration-type effect within the firm of additional coworkers on inventor productivity. With a single input, the returns to scale parameter is also the output elasticity with respect to inventor labor. For a solo-based firm, this is simply α . For a team-based firm, this is $\alpha(1 + \gamma)$. Throughout, we assume $\alpha(1 + \gamma) < 1$ to ensure decreasing returns for both types of firms (and thus an interior solution for labor demand). Our empirical estimates of the firm-level idea

production function in Section 5.1 are consistent with this assumption.

Finally, firms learn from each other in a way that affects the productivity of all firms symmetrically through the knowledge spillover ζ . This is motivated by the finding that the social returns to innovation are substantially larger than the private returns (Bloom, Schankerman, and Van Reenen 2013). We formalize spillovers in the model in the same way as Chikis, Kleinman, and Prato (2025). In particular, the spillovers generated by a particular firm come from a mix of its productivity ε_f and its size n_f and are cumulated across firms as follows:

$$\zeta = \left(\int_{f \in F} \varepsilon_f n_f^\beta df \right)^\theta, \quad (3.5)$$

where F is the measure of active firms. $\theta > 0$ governs the importance of knowledge spillovers for firm output, and $\beta \in [0, 1]$ governs how much firm size (n_f) matters compared to firm productivity (ε_f) for generating spillovers. If $\beta = 0$, the allocation of workers also does not matter because all that matters is the average productivity of firms regardless of their size. For intermediate cases, β governs the concavity of the spillover in labor, with more concavity (lower β) implying that the marginal social returns to additional inventors in a given firm are low.⁹

3.3 Firm Problem

Given their idiosyncratic productivity draw, innovative firms solve a static profit maximization problem. Producing requires a fixed operating cost wc (overhead inventor labor needed to run a lab). These firms simultaneously choose whether to produce or exit, and conditional on operating, choose the number of inventors to hire and whether to organize these inventors into solo or team production. The following characterizes the firm's profit function:

⁹Chikis, Kleinman, and Prato (2025) use data on citations patterns to the same firm in different locations where the firm has plants of varying sizes and estimate $\beta = 0.25$. In Section 5.2, we estimate β following Chikis, Kleinman, and Prato (2025) with our data.

$$\Pi(\varepsilon_f) = \max \left\{ 0, \Pi^{solo}(\varepsilon_f), \Pi^{team}(\varepsilon_f) \right\}, \quad (3.6)$$

where

$$\Pi^{solo}(\varepsilon_f) = \max_{n_f^{solo}} \zeta \varepsilon_f z^\alpha (n_f^{solo})^\alpha - w n_f^{solo} - w c, \quad (3.7)$$

$$\Pi^{team}(\varepsilon_f) = \max_{n_f^{team}} \zeta \varepsilon_f (x \Gamma(1 - \gamma))^\alpha (n_f^{team})^{(\gamma+1)\alpha} - w n_f^{team} - w c. \quad (3.8)$$

Throughout, we normalize the “price” of innovative output to 1. The model could be extended to include a production sector where ideas generated by the firm’s R&D lab are used to produce physical output.

Profit maximization creates a cutoff for ε_f above which firms choose team production. Let $\tilde{\varepsilon}^{team}$ denote this cutoff. The fixed cost creates another cutoff for firm productivity, $\tilde{\varepsilon}^{exit}$, below which firms endogenously exit. Let $d = \{0, 1\}$ denote the exit choice, which equals to 1 if firm exits.

The optimal inventor employment size of firms under solo and team production can be derived as follows:

$$n^*(\varepsilon_f) = \begin{cases} \left(\frac{\alpha \zeta \varepsilon_f z^\alpha}{w} \right)^{\frac{1}{1-\alpha}} & \text{w/ solo (if } \tilde{\varepsilon}^{exit} < \varepsilon_f < \tilde{\varepsilon}^{team} \text{)} \\ \left(\frac{(\gamma+1)\alpha \zeta \varepsilon_f (x \Gamma(1-\gamma))^\alpha}{w} \right)^{\frac{1}{1-(\gamma+1)\alpha}} & \text{w/ team (if } \tilde{\varepsilon}^{team} < \varepsilon_f \text{),} \end{cases} \quad (3.9)$$

where $\tilde{\varepsilon}^{exit}$ and $\tilde{\varepsilon}^{team}$ are determined by the following equations, respectively,

$$\tilde{\varepsilon}^{exit} = \frac{w c^{1-\alpha}}{\zeta (1-\alpha)^{(1-\alpha)} (\alpha z)^\alpha},$$

$$\tilde{\varepsilon}^{team} = \left(\frac{(1-\alpha) (\alpha z)^{\frac{\alpha}{1-\alpha}}}{(1-(\gamma+1)\alpha) ((\gamma+1)\alpha)^{\frac{(\gamma+1)\alpha}{1-(\gamma+1)\alpha}} (x \Gamma(1-\gamma))^{\frac{\alpha}{1-(\gamma+1)\alpha}}} \right)^{\frac{(1-\alpha)}{\alpha} \frac{1}{\gamma} (1-(\gamma+1)\alpha)} \frac{w}{\zeta}.$$

3.4 Firm Entry

A large mass of potential entrants can enter by paying an entry cost $w\kappa$ (innovative entrepreneurial labor) before drawing their productivity. The following free entry condition must hold in equilibrium:

$$\int \Pi(\varepsilon_f) dG(\varepsilon_f) - w\kappa = 0, \quad (3.10)$$

which pins down the mass of firms M (including potential entry). Entrants draw their idiosyncratic productivities from the same distribution g of incumbent productivities.

3.5 Labor Market Clearing

There is a fixed supply of inventors with measure 1. Labor market clearing requires that the aggregate inventor labor demand is equal to the supply:

$$M \left(\int_{\tilde{\varepsilon}^{team}} (n^{team}(\varepsilon_f; w) + c) dG(\varepsilon_f) + \int_{\tilde{\varepsilon}^{exit}}^{\tilde{\varepsilon}^{team}} (n^{solo}(\varepsilon_f; w) + c) dG(\varepsilon_f) + \kappa \right) = 1. \quad (3.11)$$

This pins down the equilibrium wage w in the economy.

3.6 Equilibrium

An equilibrium is a collection of $\left\{ \{n_f^{solo}, n_f^{team}, d_f, \Pi_f\}_f, \tilde{\varepsilon}^{exit}, \tilde{\varepsilon}^{team}, w, M \right\}$, where firms solve (3.6)-(3.9); the free-entry condition (3.10) holds; and the labor market clearing condition (3.11) is satisfied.

4 Social Planner's Problem

We next consider how the efficient allocation of inventors across firms differs from the allocation in competitive equilibrium. In this section, for expositional simplicity, we

focus on a simple case where firms operate exclusively under either solo or team production and abstract from firm entry and exit by assuming a fixed measure one of firms. This setting provides intuition for the differences between the efficient allocation and the competitive equilibrium under solo and team production and how the magnitude of misallocation varies across the two production schemes. In the quantitative model (Section 6.3), we solve the full planner's problem, which jointly determines firm size n_f , the choice between solo and team production for each firm, the mass of firms M , and the firm exit cutoff (equivalently, the mass of active firms).

We first characterize the wedges between social and private marginal product of inventors across firms with different productivity levels in each type of production (solo or team). We then compare the magnitudes of the wedges across solo and team production.

The planner allocates inventors across firms to maximize total output, internalizing spillovers and the resource constraint. The planner's problem can be written as follows:

$$\max_{n_f} Y = \int_0^1 y_f df, \quad (4.1)$$

subject to (3.4), (3.5), and

$$\int_0^1 n_f df = 1. \quad (4.2)$$

We define a firm-level wedge τ_f as the gap between the social marginal product of inventor labor at firm f (MP_f^{SP}) and the private marginal revenue product (MP_f^{CE}) perceived by the firm:

$$MP_f^{SP} = (1 + \tau_f) MP_f^{CE}. \quad (4.3)$$

4.1 Solo Production

We first solve for the optimal allocation of inventors in the case where all firms operate under solo production.

Under solo production, the planner faces the following marginal product of labor:

$$MP_f^{solo,SP} = \theta\beta\zeta^{-\frac{1}{\theta}}\varepsilon_f n_f^{\beta-1}Y + \alpha\zeta z^\alpha \varepsilon_f n_f^{\alpha-1}, \quad (4.4)$$

and the counterpart in competitive equilibrium with all solo production is

$$MP_f^{solo,CE} = \alpha\zeta z^\alpha \varepsilon_f n_f^{\alpha-1}. \quad (4.5)$$

Using (4.3), (4.4) and (4.5), the wedge under solo production is:

$$\tau_f^{solo} = \frac{\theta\beta Y}{\alpha\zeta^{1+\frac{1}{\theta}}} z^{-\alpha} n_f^{\beta-\alpha}. \quad (4.6)$$

The wedge includes a common component for all firms and a firm-specific component that depends on idiosyncratic productivity ε_f . In the spirit of [Hsieh and Klenow \(2009\)](#), we evaluate this wedge at the competitive equilibrium firm size distribution and obtain the following result.

Proposition 1. *If $0 < \beta < \alpha < 1$, firm-level wedges are decreasing in firm productivity ε_f and size n_f under solo production.*

Proof. See Appendix [A.5.1](#). □

Proposition 1 implies misallocation of inventor labor across firms. All firms are inefficiently small because of knowledge spillovers, but the planner values additional inventors at small firms the most.

4.2 Team Production

Under team production, the planner faces the following marginal product of labor:

$$MP_f^{team,SP} = \theta\beta\zeta^{-\frac{1}{\theta}}\varepsilon_f n_f^{\beta-1}Y + \alpha(1+\gamma)\zeta (x\Gamma(1-\gamma))^\alpha \varepsilon_f n_f^{\alpha(1+\gamma)-1}, \quad (4.7)$$

and the counterpart in competitive equilibrium is

$$MP_f^{team, CE} = \alpha(1 + \gamma)\zeta (x\Gamma(1 - \gamma))^\alpha \varepsilon_f n_f^{\alpha(1+\gamma)-1}. \quad (4.8)$$

Using (4.3), (4.7) and (4.8), we get the general expression for the wedge under team production as follows:

$$\tau_f^{team} = \frac{\theta\beta Y}{\alpha(1 + \gamma)\zeta^{1+\frac{1}{\theta}}} (x\Gamma(1 - \gamma))^{-\alpha} n_f^{\beta-\alpha(1+\gamma)}. \quad (4.9)$$

The following proposition shows the evaluation of the wedges at the competitive equilibrium firm size distribution under team production.

Proposition 2. *If $0 < \beta < \alpha(1 + \gamma) < 1$, firm-level wedges are decreasing in firm productivity ε_f and size n_f under team production.*

Proof. See Appendix A.5.2. □

Proposition 2 establishes that team production generates misallocation of inventor labor across firms. The following proposition shows that team production also increases the dispersion of wedges as a function of firm productivity. By increasing firms' private returns to scale, team formation causes wedges between social and private returns to decline more rapidly with firm productivity, thereby increasing the relative distortion between small and large firms.

Proposition 3. *Suppose $0 < \beta < \alpha < \alpha(1 + \gamma) < 1$. Then the wedges between social and private returns are more sensitive to firm productivity under team production than under solo production. In particular,*

$$\left| \frac{\partial \log \tau_f^{team}}{\partial \log \varepsilon_f} \right| > \left| \frac{\partial \log \tau_f^{solo}}{\partial \log \varepsilon_f} \right|.$$

Equivalently, wedges decline more rapidly with firm productivity under team production than under solo production.

Proof. See Appendix A.5.3. □

This proposition provides intuition for the result in Section 6.3 that an increase in team production increases misallocation by increasing wedges for small firms.

5 Calibration

We calibrate two sets of model parameters, one for an early period (1976-1999) and the other for a late period (2000-2018) of our sample. Several parameters are obtained by directly estimating the model's idea production function using firm-level data on patent output, inventor employment, and the within-firm share of team-based patenting compared to solo patenting. The knowledge spillover function is also directly estimated for the two periods. Other parameters are internally calibrated to match the early-period team patent share, the entry rate into patenting, and the average firm size (in terms of number of inventors). The remaining parameters are externally determined, either through normalization or taken from the literature.

5.1 Estimating the Idea Production Function

First, to pin down α , the baseline returns to scale in inventor labor under solo production, γ , the effect of team-based production on returns to scale, x , the minimum of the team productivity distribution, and the solo productivity parameter z , we estimate the model's firm-level idea production function in the data.

Specifically, we implement the following triple-interaction regression using the universe of patenting firms in the USPTO data over the period 1976–2018:

$$\log(pats_{ft}) = \mu_f + \alpha \log(n_{ft}) + \gamma^{pre}(\log n_{ft} \times \mathbb{I}(team_{ft})) + \gamma^{post}(\log n_{ft} \times \mathbb{I}(team_{ft}) \times Post_t) + \nu_{ft}. \quad (5.1)$$

Our baseline measure of firm output is patent counts, where $pats_{ft}$ denotes the num-

ber of patents filed by firm f in year t .¹⁰ The term μ_f captures a firm fixed effect, corresponding to $\log(\varepsilon_f)$ in the model, and n_{ft} is the number of (patenting) inventors employed by firm f in year t . The indicator $\mathbb{I}(team_{ft})$ equals one if the firm's share of team patents in its total patent count $\left(\frac{\text{team patents}_{ft}}{\text{total patents}_{ft}}\right)$ exceeds the sample median (0.88) in year t , and zero otherwise. $Post_t$ is a dummy equal to one for the period 2000-2018. All base effects and interactions of these regressors are included in the regression in addition to the terms in equation (5.1).¹¹

The results are reported in Column 1 in Table 1 and the resulting structural parameter estimates are summarized in Table 3. We find evidence of decreasing returns to scale in inventor labor in the baseline period (pre-2000), with an estimate of $\alpha = 0.757 < 1$ and no evidence of a change in returns to scale for solo firms over time (the coefficient on $\log(n_{ft}) \times Post_t$ is close to zero and not statistically significant).

Consistent with the model, team production increases returns to scale with the estimate of $\gamma^{pre} = \frac{0.039}{\alpha}$. This implies $\gamma = 0.052$ in the early period. Together the estimates of α and γ imply that team based firms still have decreasing returns to scale, but less-decreasing than solo-based firms where inventors tend to work alone (around 0.796 instead of 0.757 for the early period).

The positive coefficient on the triple interaction indicates that the returns to scale benefit of teams γ rises in the later period (post-2000), implying a thicker right tail of pairwise productivity draws, possibly due to increasing knowledge specialization. Consistent with the idea of rising complementarities across inventors over time, we document in Appendix C.4 that patenting teams increasingly combine inventors from different technological subfields, defined by the technology class in which an inventor patents most frequently. Quantitatively, the late-period estimate is $\gamma = \frac{0.0394+0.0312}{\alpha} =$

¹⁰To enter the data, a firm-year observation must have at least one patent, so taking logs does not create a problem in that dimension.

¹¹Identification relies on variation in the choice of using solo and team production, conditional on firm fixed effects μ_f and inventor employment size n_{ft} . A potential concern is that this choice may be correlated with unobserved determinants of output. We are exploring the use of various instruments for this choice including the past teamwork experience of new inventors at the firm and communication costs within the firm (Bloom et al. 2014; Jiang 2024).

0.093.

To infer solo inventor productivity z from the regressions, we first need to partial out the effect of knowledge spillovers ζ , where $\zeta^{pre} = \left(\int_{f \in F} \varepsilon_f n_f^{\beta^{pre}} df \right)^\theta$, where β^{pre} comes from 5.2. z^{pre} is set to $z^{pre} = \left(\frac{\exp(0.065)}{\zeta^{pre}} \right)^{\frac{1}{\alpha}} = 1.247$, which comes from the mapping $constant = \alpha \ln z^{pre} + \ln \zeta^{pre} = 0.065$. The early period value of minimum team productivity x is internally calibrated to match the team patent share in the early period.

Finally, the negative coefficients associated with the dummies $Post_t$ and $Post_t \times \mathbb{I}(team_{ft})$ indicate a general decline in patenting productivity across firms in the later part of the sample period. This pattern may reflect a broader decline in research productivity, in line with the evidence that ideas have become harder to find (Bloom et al. 2017). Based on these, we calibrate z^{post} and x^{post} for the late period. In particular, we use the constant and the coefficient on $Post_t$ to set z^{post} to $z^{post} = \left(\frac{\exp(0.065 - 0.019)}{\zeta^{post}} \right)^{\frac{1}{\alpha}} = 1.374$ where $\zeta^{post} = \left(\int_{f \in F} \varepsilon_f n_f^{\beta^{post}} df \right)^\theta$. In a similar fashion, we set x^{pre} to match with the share of team patents in the early period, and use the coefficient on $Post_t \times \mathbb{I}(team_{ft})$ to infer the changes in minimum team productivity between the two periods. And then, we use the mapping $x^{post} = \frac{\left(\frac{\zeta^{pre}}{\zeta^{post}} \exp(-0.049) \right)^{\frac{1}{\alpha^{post}}}}{\Gamma(1 - \frac{1}{\gamma^{post}})} \left(x^{pre} \Gamma(1 - \frac{1}{\gamma^{pre}}) \right)$, which captures the estimated change in team productivity between the early and late periods. This pins down $x^{post} = 1.096$ for the late period. Note that, as spillovers decline in the late period—capturing the observed decline in firm productivity—the model calibrates higher values of z and x to match the data. This suggests that inventors themselves are not inherently less productive and that the decline in patent productivity was instead driven purely by the decline in knowledge spillovers across firms.

Robustness. Table 1, Column 2 report estimates from the same specification, limiting the sample to public firms, which allows us to control for capital inputs into R&D activities using the Peters and Taylor (2017) measures of intangible capital (capitalized selling and general administrative and R&D expenses). Controlling for these inputs

Table 1: Returns to Scale in Team Production

	(1)	(2)
	All	Public Firms
$\log(n_{ft})$	0.757*** (0.002)	0.818*** (0.004)
$\log(n_{ft}) \times \text{Team}_{ft}$	0.039*** (0.004)	0.0875*** (0.006)
$\log(n_{ft}) \times \text{Team}_{ft} \times \text{Post}_t$	0.031*** (0.004)	0.011* (0.006)
Team_{ft}	-0.677*** (0.004)	-0.732*** (0.008)
Post_t	-0.019*** (0.003)	-0.058*** (0.009)
$\log(n_{ft}) \times \text{Post}_t$	0.003 (0.002)	0.019*** (0.004)
$\text{Team}_{ft} \times \text{Post}_t$	-0.049*** (0.005)	-0.015 (0.011)
$\log(K_{ft}^{intan.})$		-0.012*** (0.002)
Constant	0.065*** (0.002)	0.081*** (0.014)
R-squared	0.704	0.817
Obs.	1105437	131166
Firm FE	Yes	Yes

Notes: Source: USPTO, 1976–2018. Column (1) includes all firm-year observations. Column (2) includes public firms with available intangible capital data from [Peters and Taylor \(2017\)](#). See equation (5.1) for details. Standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

addresses concerns that omitted inputs (capital) that are positively correlated with inventor employment or team usage could bias the estimated returns to scale upward. Across both subsamples, our main findings remain qualitatively robust.

In Appendix Table C4, we re-estimate equation (5.1) using forward citations instead of patent counts as the measure of firm output, adjusted for technology-class fixed effects for citations. In this case, the returns-to-scale advantage of team production is statistically significant only in the post-2000 period but the general patterns remain similar to the baseline results.

Table 2: Returns to Scale in Knowledge Spillovers

	(1)	(2)
$\log(n_{fmt})$	0.429*** (0.011)	0.467*** (0.015)
$\log(n_{fmt}) \times \text{Post}_t$		-0.061*** (0.016)
R-squared	0.147	0.148
Obs.	106,362	106,362
Firm-Year FE	Yes	Yes

Notes: The dependent variable is the log number of external citations received by firm f 's patents from other firms in market n (commuting zone) within a five-year window. The key explanatory variable is the log number of inventors at firm f in market n . Column (2) allows the elasticity of knowledge spillovers with respect to inventor employment to differ after 2000 through an interaction with Post_t . All specifications include firm-year fixed effects. See (5.2) for details. Standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Another concern is that inventor employment is measured with error, since the USPTO data capture only inventors who successfully patent in a given year. To address this issue, we use Revelio Labs data on inventor employment histories to construct a better measure of inventors per firm. However, these data are reliable only from around 2008 onward. Appendix Table C5 reports the corresponding estimates. While the baseline returns-to-scale parameter (α) is lower in this sample, the estimated team effects (γ) in the pre- and post-periods (2008–2014 versus 2014–2018) remain qualitatively consistent with our main findings and in fact the estimated returns to scale benefit from using teams is much larger in this sample.

5.2 Estimating the Knowledge Spillover Function

We use local citation flows to estimate the concavity of the model’s knowledge spillover function in firm size (β). We estimate the following regression:

$$\log(\iota_{fmt}) = \mu_{ft} + \beta_{pre} \log n_{fmt} + \beta_{post} \log(n_{fmt} \times Post_t) + \varepsilon_{fmt}, \quad (5.2)$$

where ι_{fmt} is the number of five-year forward external citations that firm f ’s patents in year t receive from other firms in market m (commuting zone) divided by the number of firms in market m in year t , n_{fmt} is the number of inventors at firm f in market n , μ_{ft} is a firm-year fixed effect, and $Post_t$ is an indicator for the post-2000 period.

Our approach is therefore identical to (Chikis, Kleinman, and Prato 2025), except that we allow β to vary between the two time periods since our focus is on changes in misallocation over time.¹² The coefficient β_{pre} associated with the number of inventors captures the returns to scale of R&D labor in creating spillovers for the early period, and the coefficient β_{post} associated with the interaction term measures the change in these returns during the post-2000 period.

Table 2 presents the estimation results. Column 1 estimates the returns to scale in knowledge spillovers using the pooled sample and yields an estimate of $\beta = 0.429$. Column 2 allows it to differ between the pre- and post-2000 periods. The estimates imply that the returns to scale of spillovers with respect to firm size were $\beta = 0.467$ before 2000 and declined to $\beta = 0.406$ after 2000. Allowing for differences in β over time is important for separating in the model how much of the decline in knowledge spillovers and the increase in misallocation is due to changes in the idea production function that favor teams versus a technological change in the knowledge spillover function that would have reduced spillovers and increased misallocation without any changes in the

¹²While the main data (USPTO) is the same in both studies, Chikis, Kleinman, and Prato (2025) use Dun and Bradstreet data to identify establishments whereas we rely on inventor location in patent data as a proxy for establishment location.

Table 3: Calibration

Panel A: Early Period (1976-1999)			
Parameter	Definition	Value	Target/Source
α	RTS, solo firms	0.757	Table 1
γ	Higher RTS, team firms	0.052	Table 1
z	Solo prod.	1.247	Table 1
x	Location for team prod.	1.065	Share of team patents
$x_{\min}^f$	Location for ε	1	Normalization
η	Tail for ε	0.164	Emp. share of large firms
β	scale for n in spillovers	0.467	Table 2
θ	overall spillovers	0.080	Chikis, Kleinman, and Prato (2025)
κ	entry cost	1.456	Patenting entry rate
c	fixed cost	0.512	Average firm size
Panel B: Late Period (2000-2018)			
Parameter	Definition	Value	Target
α	RTS, solo firms	0.757	Table 1
γ	Higher RTS, team firms	0.093	Table 1
z	Solo prod.	1.374	Table 1
x	Location for team prod.	1.096	Table 1
$x_{\min}^f$	Location for ε	1	Normalization
η	Tail for ε	0.164	Fixed to early period
β	scale for n in spillovers	0.406	Table 2
θ	overall spillovers	0.080	Chikis, Kleinman, and Prato (2025)
κ	entry cost	1.456	Fixed to early period
c	fixed cost	0.512	Fixed to early period

Notes: This table reports the parameter values for the calibrated model in the early (Panel A) and late period (Panel B) and the target or source for each parameter value.

idea production function.

5.3 Other Parameters: Early Period

Based on the estimation results, we calibrate the model as follows. Table 3, Panel A reports the calibrated parameters for the early period. We first choose a functional form for the exogenous firm productivity distribution G : a Pareto distribution with shape $x_{\min}^f$ and scale η . Then we externally calibrate six parameters, $(\alpha, \gamma, z, x_{\min}^f, \beta, \theta)$ as follows. We use the estimation results in Table 1 to pin down the returns to scale in

production, α , the returns to scale in team productivity, γ , and the productivity of solo inventors, z . We normalize the lower bound of the firm-level productivity distribution, $x_{\min}^f$, to one. We calibrate β using the estimates reported in Table 2. For θ , we use the estimate from Chikis, Kleinman, and Prato (2025), who identify values in the range $[0.04, 0.12]$.¹³

Next, we internally calibrate four parameters, $(x_{\min}^{team}, \eta, \kappa, c)$, to match four empirical moments. The fit for these targets is reported in Table 4. First, we target the share of team patents over 1976–1999, equal to 57.8 percent (see Figure 1), which pins down the lower bound of team productivity, x . Second, the Pareto tail parameter of the firm productivity distribution, η , is chosen to match the employment share of large firms (with 145 inventors or more), which was 32.8% in the early period (Figure 2). Third, we target the entry rate of patenting firms, defined as the share of entrants (age 0) among all patenting firms using Business Dynamics Statistics (BDS) data over 1978–1999, which yields an average annual entry rate of 2.7 percent.¹⁴ This moment pins down the entry cost, κ . Finally, we choose the operating fixed cost, c , to match the average inventor employment size of patenting firms in the USPTO data over 1976–1999, equal to 7.1 inventors.

5.4 Other Parameters: Late Period

Our central question is whether changes in the idea production function can account for the observed rise in inventor employment concentration. We hold fixed the parameters related to industry primitives (the productivity distribution parameter η , the entry cost κ and the fixed cost c) and vary the parameters related to the idea production function and the knowledge spillover function using the directly estimated values from equation (5.1). Panel B of Table 3 reports the parameter values for the late period.

¹³This range is also consistent with estimates reported in (Greenstone, Hornbeck, and Moretti 2010; Moretti 2021) using different samples and time periods.

¹⁴The BDS data are available starting in 1978.

Table 4: Targeted Moments For the Early Period (1976-1999)

Target Moment	Data	Model
Share of team patents	57.8	57.8
Employment share of firms with $n > 145$	32.8	32.8
Patenting entry rate (annual)	2.6	2.6
Average firm size (inventors)	7.1	7.1

Notes: This table reports the targeted moments in the data (in the first column) and their corresponding values implied by the model (in the second column). The data sources are the USPTO (1976-1999) and the U.S. Census Bureau Business Dynamics Statistics (BDS, 1978-1999).

6 Quantitative Results

This section first describes the model fit for targeted and non-targeted moments, then considers various counterfactuals to decompose the role of the idea production parameters and the knowledge spillover parameter in explaining changes in the firm size distribution, spillovers across firms, and output over time. The final part of this section compares the laissez faire economies in the early and late period to their efficient counterparts and shows that misallocation has risen over time due to changes in the idea production function that favor teams.

6.1 Model Fit and Aggregate Implications

Table 4 shows that the calibrated model exactly matches the targeted moments for the early period. Table 5 reports the model predicted levels of and changes in various aggregate moments over time due to the estimated changes in the idea production function, including various non-targeted moments.

The model predicts a 23.4% increase in inventor wages, consistent with the evidence in Ekerdt and Wu (2025), who document a rise of roughly 15 percentage points in the wage premium for researchers since 1970. The increase in the inventor wage increases both the equilibrium exit cutoff and the cutoff for using teams.

The model implies a decline in the share of firms operating with solo production

Table 5: Model Predicted Changes in Aggregate Moments

	Early	Late	Change
Inventor wage w	0.724	0.894	23.4
Prod. cutoff (exit)	1.004	1.263	25.8
Prod. cutoff (team)	1.580	1.657	4.9
Share of solo-based firms	93.7	80.9	-12.8
Share of team patents	57.8	92.7	34.9
Employment share of firms with $n > 145$	32.8	79.6	46.7
Average firm size	7.1	30.2	325.8
Spillovers ζ	0.903	0.823	-8.8
Output (normalized)	1.000	1.234	23.4

Notes: This table reports the model-implied changes in aggregate moments between the early and late periods. The first two columns are the moments for the early and late periods, respectively, and the third column reports the corresponding percent or percentage point change relative to the early-period benchmark.

of 12.8 percentage points. Consistent with this prediction, the data show a decline of approximately 12 percentage points in the share of innovative firms with only one inventor (from 39% to 27% of firm–year observations). Using our empirical definition of solo-based firms from the regression (those with below-median team patent share), the share of solo-based firms declines by 15 percentage points, from 60% to 45%. Average firm size increases from 7.1 inventors to 30.2, whereas in the data the average size rose to 10.8 inventors per firm. The model also predicts a substantial rise in employment concentration due to changes in the idea production function. The employment share of large firms (defined as those with 145 or more inventors) increases from 32.8% to 79.6%, compared with an increase to 44.1% in the data.¹⁵

Together, these changes reduce knowledge spillovers by 8.8%. This suggests a possible microfoundation for the declining knowledge diffusion hypothesis of [Akcigit and Ates \(2023\)](#): shifts in the idea production function generate greater misallocation of inventors across firms by concentrating inventors in large firms, which in turn dampens spillovers since the returns to additional inventors in large firms in terms of additional

¹⁵Appendix B.1 reports two additional experiments where the exogenous firm productivity distribution is allowed to vary, either jointly with the idea production parameters, or on its own, to match the rise in employment concentration.

knowledge spillovers are low. However, the rise in returns to scale generated by team production increases output despite the decline in spillovers across firms. The next subsection decomposes the role of individual idea production parameters in more detail and we return to the issue of declining knowledge spillovers and misallocation in Section 6.3.

Finally, as an additional validation of the model’s key prediction that team patent share and concentration are correlated in the aggregate, we also investigate this correlation at the technology class (CPC group) level in Appendix C.3. Inventor concentration is positively associated with the share of team patents at the CPC technology-class level in the cross section (Table C6) and changes in the team patent share within a CPC group over time are positively associated with an increase in inventor concentration (Figure C7).

6.2 Decomposition of Aggregate Outcomes

This section decomposes the contribution of each estimated change in the idea production function parameters (z, x, γ) individually and isolates the role of idea production function changes from the role of spillovers to the evolution of key aggregate moments: the share of team patents, employment concentration, as well as innovative output and knowledge spillovers. This exercise isolates the role of each parameter shift and clarifies the mechanisms through which they affect the aggregate outcomes.

Table 6 shows the results for each of the idea production parameters one at a time. Table 7 presents the results for the changes of all three idea production parameters (“IP only”) with all else (including β) fixed to the early period. Below, we discuss the effects of the parameter changes for each aggregate variable in turn.

Share of team patents. The increase in solo productivity z on its own reduces the share of team patents because it benefits solo production relative to team production. On the other hand, the increase in the minimum team productivity x can explain about

Table 6: Decomposing the Role of Idea Production Function Changes

	Early	Late	z only	Change
Inventor wage w	0.724	0.894	0.760	4.9
Prod. cutoff (exit)	1.004	1.263	1.000	-0.4
Prod. cutoff (team)	1.580	1.657	2.422	53.2
Share of solo-based firms	93.7	80.9	99.5	5.9
Share of team patents	57.8	92.7	29.5	-28.3
Employment share of firms with $n > 145$	32.8	79.6	26.8	-6.0
Average firm size	7.1	30.2	6.5	-8.0
Spillovers ζ	0.903	0.823	0.908	0.6
Output (normalized)	1.000	1.234	1.049	4.9
	Early	Late	x only	Change
Inventor wage w	0.724	0.894	0.760	1.1
Prod. cutoff (exit)	1.004	1.263	1.018	-0.4
Prod. cutoff (team)	1.580	1.657	1.399	53.2
Share of solo-based firms	93.7	80.9	85.6	-8.1
Share of team patents	57.8	92.7	69.4	11.6
Employment share of firms with $n > 145$	32.8	79.6	34.2	1.4
Average firm size	7.1	30.2	7.8	9.4
Spillovers ζ	0.903	0.823	0.900	-0.3
Output (normalized)	1.000	1.234	1.011	1.1
	Early	Late	γ only	Change
Inventor wage w	0.724	0.894	0.876	21.0
Prod. cutoff (exit)	1.004	1.263	1.327	32.1
Prod. cutoff (team)	1.580	1.657	1.454	-8.0
Share of solo-based firms	93.7	80.9	42.8	-50.9
Share of team patents	57.8	92.7	97.8	40.1
Employment share of firms with $n > 145$	32.8	79.6	80.4	47.6
Average firm size	7.1	30.2	41.0	477.7
Spillovers ζ	0.903	0.823	0.827	-8.4
Output (normalized)	1.000	1.234	1.210	21.0

Notes: This table reports the counterfactual exercises in which we vary one parameter at a time— z (top panel), x (middle panel), and γ (bottom panel)—while holding all other parameters fixed at their early-period values. The first two columns are the model moments for the early and late periods, respectively. The third column presents the model-implied value when only the given parameter is adjusted to its late-period estimate, and the last column reports the corresponding percent or percentage change relative to the early-period benchmark.

a third of the model-predicted increase in the share of team patents. The increase in the returns to scale benefit of teams, captured by the thicker right tail of the team productivity distribution (higher γ), increases the predicted team patent share substantially,

Table 7: Decomposing the Role of Idea Function vs. Spillover Changes

	Early	Late	IP only	Change
Inventor wage w	0.724	0.894	0.899	24.1
Prod. cutoff (exit)	1.004	1.263	1.270	26.4
Prod. cutoff (team)	1.580	1.657	1.633	3.3
Share of solo-based firms	93.7	80.9	78.4	-15.3
Share of team patents	57.8	92.7	93.3	35.5
Employment share of firms with $n > 145$	32.8	79.6	79.7	46.9
Average firm size	7.1	30.2	31.2	340.2
Spillovers ζ	0.903	0.823	0.833	-7.7
Output (normalized)	1.000	1.234	1.241	24.1

Notes: This table reports the counterfactual exercises in which we vary the three parameters (z, x, γ) in idea production function while holding all other parameters fixed, including β , at their early-period values. The first two columns are the model moments for the early and late periods, respectively. The third column presents the model-implied value when only the three parameters are adjusted to its late-period estimate, and the last column reports the corresponding percent or percentage change relative to the early-period benchmark.

to 97.8% of all patents. The joint changes in the idea production parameters explain the entire rise in the team patent share between the early and late periods of the model (Table 7).

Inventor employment concentration. An increase in solo productivity z increases the productivity cutoff for adopting team production. This in turn decreases the average firm size slightly and results in a decline in the employment share of large firms. The opposite is true for an increase in the minimum team productivity x , which increases average firm size and employment concentration slightly. The increase in the right tail of team productivity draws (i.e., higher γ) generates a substantially larger rise in average firm size and the employment share of large firms. The change in employment concentration between the early and late periods in the model is entirely driven by the idea production parameters (Table 7).

Innovative output and knowledge spillovers. The rise of solo productivity and the minimum team productivity each increase innovative output but have only minor effects on knowledge spillovers; the rise in solo productivity raises knowledge spillovers

by reallocating inventors toward smaller firms so increases output by more compared to the rise in team productivity because the rise in team productivity has countervailing negative effects on knowledge spillovers, as we discuss more in the next section. Similarly the increase in the returns to scale benefit of using teams (higher γ) raises output substantially but also reduces knowledge spillovers substantially since workers are reallocated toward larger firms. In Table 7, changes in the idea production parameters account for 88% of the decline in knowledge spillovers (suggesting the decline in β is responsible for the remaining 12%).

6.3 Effects on Misallocation

Finally, the main results compare the competitive equilibrium allocations in the early and late periods with the corresponding social optimum given the different idea production technologies in the two periods. To do this, we extend the planner's problem in Section 4 to allow the planner to choose the mass of firms, firm entry and exit, and assign firms to use solo or team production. The full set-up is outlined in Appendix A.6. The parameter values are the same as in Table 3.

Aggregate outcomes. We first quantify the gap between the competitive equilibrium and the social planner's allocation in each period to measure aggregate misallocation and then explore the mechanisms. Table 8 reports the aggregate results, measuring the deviation of the competitive equilibrium from the social optimum in aggregate innovative output, knowledge spillovers, and the employment share of large firms.

In the early period, innovative output in the competitive equilibrium is 0.20% below the efficient allocation. In the late period, the gap is nearly ten times larger (1.91%). The changes in the idea production function (absent the change in the spillover function parameter β) increase the output gap by seven times its early value, suggesting they are responsible for around 70% of the change in the output gap between the competitive equilibrium and the planner's allocation. The endogenous decline in knowledge

Table 8: Competitive Equilibrium vs. Social Planner (CE to SPP)

Variable	Early	Late	IP Only
Total Output	-0.20%	-1.91%	-1.40%
Spillovers ($z^{spillover}$)	-0.60%	-6.33%	-5.84%
Emp. share of firms with $n > 145$	7.43 pp	21.69 pp	22.46 pp

Notes: This table reports the deviation of the competitive equilibrium (CE) from the social planner (SPP) for aggregate innovative output, knowledge spillovers, and the employment share of large firms. The first two columns report the early- and late-period economies. The final column reports a counterfactual in which only idea production function parameters change from their early-period values, while the knowledge spillover parameter β remains fixed at its early-period value.

spillovers in the competitive equilibrium is a major driver of this widening gap, and the gap between the planner's realized spillovers and the competitive equilibrium's spillovers grows by a factor of ten in the late period.

The output gains under the planner's allocation arise through two channels. First, the planner internalizes knowledge spillovers and retains a larger mass of firms, and second, the planner expands low-productivity firms and assigns them to team production when socially optimal. In the competitive equilibrium, the higher equilibrium wage induces excessive exit, resulting in a smaller mass of active firms, and prevents low-productivity firms from scaling up to the point at which team productivity exceeds solo productivity. The planner corrects this distortion by reallocating inventors and lowering the productivity thresholds for adopting team production and firm exit.

Finally, the planner chooses a lower employment share of large firms than the competitive equilibrium in both periods, and this gap widens significantly in the late period. Note that, comparing the planner's allocations across time periods, the planner still chooses a much higher employment share of large firms in the late period compared to the early period in response to the team production shock favoring large firms. However, the increase in inventor concentration chosen by the planner is considerably smaller than the pronounced rise in employment concentration observed in the competitive equilibrium.

Firm size decision rules. To understand the mechanism, we compare the firm-size decision rules under the competitive equilibrium and the social planner's allocation in both periods. Figure 4 illustrates the decision rules separately for the early and late periods. To enhance visibility, we split the figure into two subfigures that zoom in on the low- and high-productivity regions, shown in the left and right panels for each period. We find differences at both the extensive and intensive margins of inventor employment, as well as in the firm exit cutoffs.

At the intensive margin, the planner reallocates inventors from larger toward smaller firms, increasing the scale of low-productivity firms and reducing the scale of high-productivity firms relative to the competitive equilibrium. This reallocation reflects the planner's internalization of knowledge spillovers, which are disproportionately generated by smaller firms. In addition, firms with productivity between the planner's and competitive equilibrium team cutoffs adopt team production only under the planner, further increasing their optimal scale. Lastly, at the extensive margin, the social planner sets the exit cutoff at the lower bound of the productivity distribution, $\tilde{\varepsilon}_{SPP}^{exit} = 1$, implying no firm entry nor exit. In contrast, the competitive equilibrium features a slightly higher exit cutoff, $\tilde{\varepsilon}_{CE}^{exit} = 1.004$, which generates a positive entry and exit rate of 2.64%. As a result, the planner sustains a larger mass of firms than in the competitive equilibrium, $M_{SPP} = 0.123 > M_{CE} = 0.113$. Here, the planner internalizes the benefits of additional firms in the spillover term unlike entrants in the competitive equilibrium. Increasing the mass of firms allows the planner to spread labor across more firms to enhance spillovers given the concavity in the spillover function.

Comparing the early and late periods, at the extensive margin, the productivity cutoff for adopting team production declines in the planner's equilibrium to take advantage of the greater productivity of teams, while it rises in the competitive equilibrium because of the increase in the wage in competitive equilibrium. At the intensive margin, conditional on firm productivity, solo firms become smaller in the late period, while team-based firms become larger relative to the early period in both the competitive equilibrium and the planner's solution. However this pattern is more pronounced in

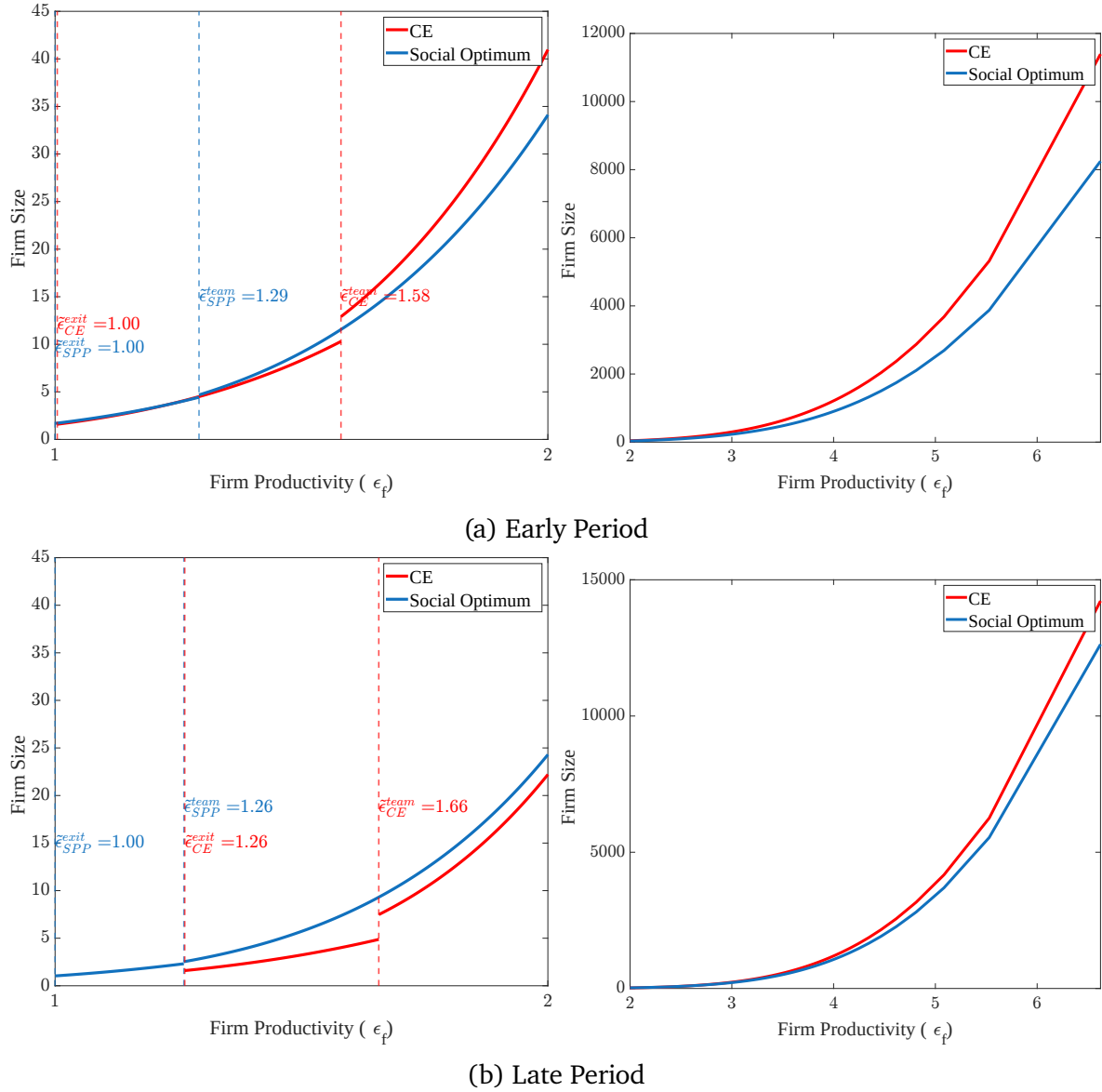


Figure 4: Firm Size Decision Rule: CE vs. SPP in the Early and Late Periods

Note: These figures illustrate firm-size decision rules under the competitive equilibrium and the social planner's allocation for the early period (top panel) and the late period (bottom panel). In each panel, the left subfigure focuses on the low-productivity (small-firm) region, while the right subfigure focuses on the high-productivity (large-firm) region to highlight the underlying patterns clearly. The red lines correspond to the competitive equilibrium, and the blue lines indicate the planner's equilibrium. The vertical lines denote the two productivity cutoffs for adopting team production and firm exit in each equilibrium for each period.

the competitive equilibrium, particularly for solo firms. Thus, the competitive equilibrium amplifies the dispersion in firm size relative to the efficient allocation in the late period. These shifts are clearly illustrated in Figure B5 in the Appendix, where we plot the same series across periods within each competitive equilibrium and the social planner's equilibrium.

Firm-level wedges. Finally, we compute the firm-level wedges τ_f , defined in (4.3) as the gap between the social marginal product and private marginal revenue product of inventor labor, evaluated at the competitive equilibrium allocation. Note that allowing an endogenous choice between solo and team production creates a productivity cutoff in both the social planner's allocation and the competitive equilibrium. Also, endogenous firm entry and exit creates another productivity cutoff that drives firms to exit. We therefore evaluate the wedges accordingly by taking into account the optimal regime of production under each case.¹⁶

Figure 5 plots the wedges as a function of firm-level productivity, ε_f . In both periods, the wedges are decreasing in firm productivity (and therefore in firm size), consistent with the analytical results in Propositions 1 and 2. In addition, within each period, there is a modest increase in wedges around the social planner's team-production cutoff, $\tilde{\varepsilon}_{SPP}^{team}$. This pattern reflects differences at both the intensive and extensive margins between the competitive equilibrium and the planner's allocation.

At the intensive margin, as seen in Figure 4, there is a gap in firm size conditional on productivity under both solo and team production. In particular, the social planner selects a larger (smaller) size for smaller (larger) firms relative to the competitive equilibrium, which creates a source of wedges.

At the extensive margin, the competitive equilibrium features a higher cutoff for

¹⁶Specifically, as in Figure 4, for firms below the social planner's team cutoff $\tilde{\varepsilon}_{SPP}^{team}$, both allocations involve solo production. For firms between the planner $\tilde{\varepsilon}_{SPP}^{team}$ and competitive equilibrium team cutoffs $\tilde{\varepsilon}_{CE}^{team}$, the planner adopts team production while the competitive equilibrium remains in solo production. For firms above the competitive equilibrium cutoff $\tilde{\varepsilon}_{CE}^{team}$, both allocations involve team production. Furthermore, there are no active firms below the competitive equilibrium exit cutoff, $\tilde{\varepsilon}_{CE}^{exit}$, unlike in the planner's allocation, so we cannot evaluate wedges for these firms.

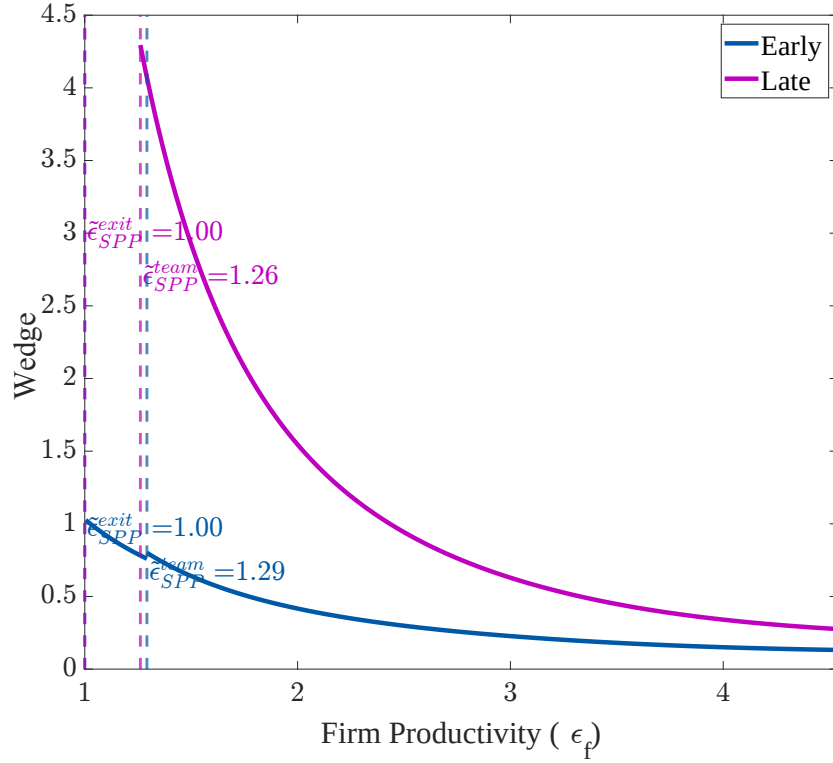


Figure 5: Firm-level Wedges: CE vs. SPP in the Early and Late Periods

Note: The figure illustrates the firm-level wedges τ_f between the competitive equilibrium and the social optimum, defined in equation (4.3), as a function of firm productivity ϵ_f . The blue lines correspond to the early period, and the purple lines correspond to the late period. The dashed vertical lines denote the two productivity cutoffs for adopting team production and firm exit in each period.

adopting team production than the planner, $\tilde{\varepsilon}_{CE}^{team} > \tilde{\varepsilon}_{SPP}^{team}$, in both periods. Consequently, there exists a productivity range between these cutoffs in which firms choose solo production in the competitive equilibrium even though team production is socially optimal. This discrepancy in organizational choice introduces an additional source of misallocation. Such a margin is absent in the simplified framework of Section 4.

Furthermore, misallocation arises from the difference in exit cutoffs between the competitive equilibrium and the planner’s allocation. The planner choose a lower exit cutoff and thus assigns inventors to firms over a wider range of productivity, whereas firms exit earlier in the competitive equilibrium. As a result, there exists a range of productivity below the competitive equilibrium exit cutoff, $\tilde{\varepsilon}_{CE}^{exit}$, in which no firms operate under the competitive equilibrium but remain active under the planner. Because the competitive equilibrium assigns zero scale to these firms, wedges may become unbounded in this range. We therefore exclude this range in Figure 4 and truncate the figure at the competitive equilibrium exit cutoff in both periods.

Comparing wedges across the two periods, the level of wedges is uniformly higher in the late period across the entire productivity distribution. Moreover, the increase in wedges around the social planner’s team-production cutoff is more pronounced in the late period, resulting in a steeper slope in that productivity range. This comes from the above mechanism through intensive and extensive margin differences. Lastly, as the competitive equilibrium exit cutoff rises in the late period, amplifying misallocation along the firm exit margin. Implementation of the first best would therefore involve larger R&D subsidies to small firms and/or a higher subsidy to entry in the later period.

7 Conclusion

This paper studies how the rise of team-based idea production interacts with firm boundaries and shapes the allocation of innovative labor across firms and its aggregate implications. We document a dramatic shift toward team production in U.S. patenting and a concurrent rise in inventor employment concentration at large firms. We

also provide evidence that firms play a central role in determining the set of possible collaborators and, therefore, the boundaries of innovative teams.

To interpret these patterns, we develop a tractable general equilibrium model in which team formation occurs within firms and knowledge spillovers operate across firms. We introduce a new microfoundation for team formation that delivers higher returns to scale in inventor labor for firms that choose production in teams, influencing firm size, the allocation of inventors to firms, and aggregate innovative output and knowledge spillovers.

We estimate the model's idea production function using the U.S. patent data and show that team production increases returns to scale in inventor labor in the data, consistent with our new microfoundation. We further find that the idea production function has changed over time to increasingly favor teams and offer an explanation for this change based on increased knowledge specialization of inventors. Our model predicts that these shifts in the idea production function increase inventor employment concentration while endogenously reducing knowledge spillovers across firms.

The welfare implications of rising team production therefore depend critically on firm boundaries and the structure of knowledge spillovers. One promising direction for future research is to examine how the transition toward team-based idea production, particularly within a smaller set of innovative firms, shapes the long-run dynamics of knowledge accumulation for R&D workers.

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A Model Appendix

A.1 Empirical Support for Modeling Assumptions

One model assumption that deserves additional discussion is the assumption that there is no serial collaboration or team-specific human capital. Our model is a static, one period model, so there is no natural way of modeling accumulation of team-specific capital. Moreover, somewhat surprisingly, patenting teams are quite short lived in the data: 80% of teams in our sample only patent together once. Even for pairs of inventors within larger teams, 65% of pairs never collaborate again.

An illustrative example is inventor Adam Barth whose patenting history is as follows:

1. 2011, Google: "Automatically updating browser extensions, and applications thereof" w/ Johann Tomas Sigurdsson, Sigurdur Asgeirsson, Roger Tawa, Jeffrey Bailey
2. 2012, Google: "Method and system for browser identity" w/ Erik Kay and Aaron Boodman
3. 2013, Google: "Heterogeneous virtual machines sharing a security model" w/ Charles Reis
4. 2014, Alarm.com: "Force-sensitive occupancy sensing technology" w/ Mark Andrew Hanson, Samuel Alden Ridenour, Paul Michael Wempe
5. 2015, Alarm.com: "Fall detection and reporting technology" w/ Mark Andrew Hanson, Jean-Paul Martin, Christopher Silverman
6. 2017, Alarm.com: "Security system communicator and keypad device" w/ Zackary Watson, Daniel Todd Kerzner
7. 2020, Alarm.com: "Health and wellness management technology" w/ Mark Andrew Hanson, Christopher Silverman

At Google, a firm with a large pool of inventors, he had many distinct collaborations working on different problems. After moving to Alarm.com, a much smaller firm, he still worked with different co-inventors on different problems, but began to have some

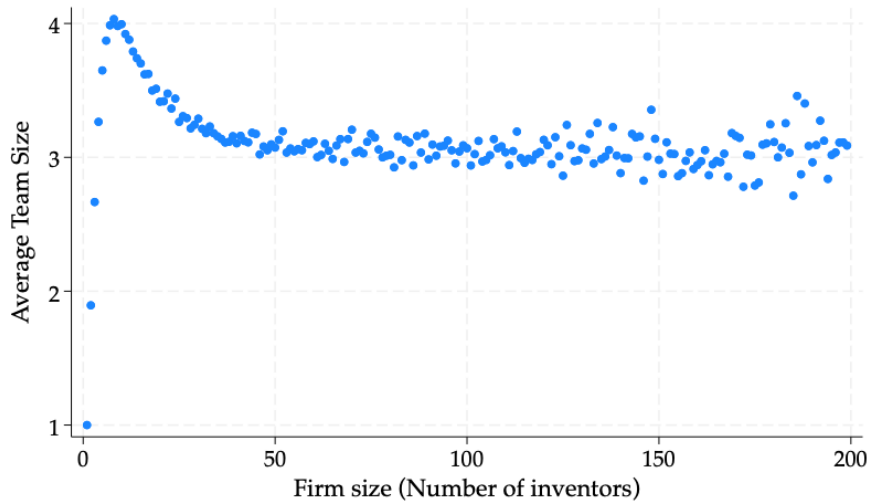


Figure A1: Average Team Size By Number of Inventors

Note: The figure illustrates the average team size within firms, as a function of the total number of patenting inventors at the firm in that year pooling all years in the USPTO data, 1976-2023.

repeated collaborations.

A second assumption is that inventors only work in teams of two. This assumption is stylized, since we also observe many teams of three to five inventors in the data (teams of two constitute 37% of team patents, and teams of less than 6 team members together constitute 80%). We expect a similar increasing, concave relationship between inventor productivity and the number of coworkers would hold in a richer model with productivity draws for larger groups of inventors. Importantly, the number of inventors per patent is not increasing one for one in the number of inventors at the firm (that is, firms with many inventors still divide them into smaller teams rather than pooling them into a single large team), leveling off at around three inventors (Figure A1).

In partial equilibrium the model predicts that output per worker is increasing in the number of coworkers. Figure A2, which plots patenting output per worker as a function of the number of (successful) inventors, shows that this is the case separately for two subperiods using USPTO data. Consistent with our finding that returns to scale increase, there is a level shift downward in output per worker in the later period. One concern with the figure is that output per worker looks highest for firms with one in-

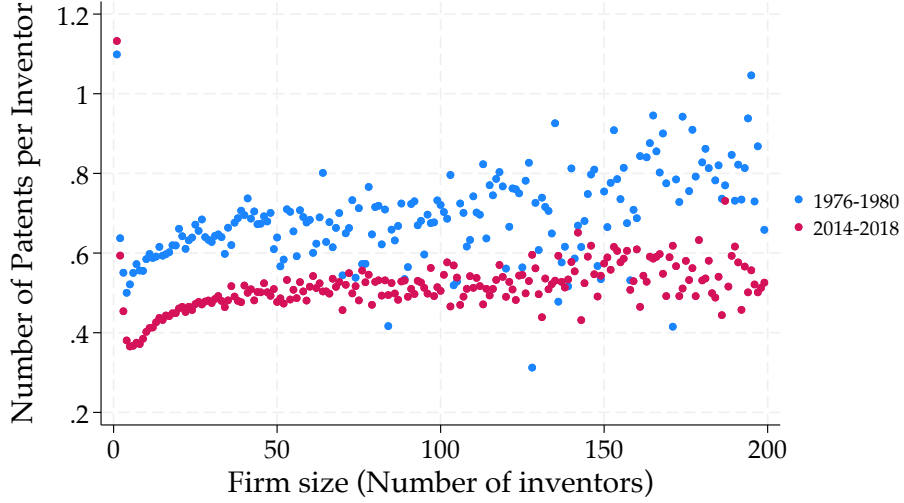


Figure A2: Number of patents per worker as a function of firm size in the USPTO data for two subperiods: 1976-1980 and 2014-2018.

Note: The figure illustrates the number of patents per worker within firms, as a function of the total number of patenting inventors at the firm in that year. The blue dots pool all years in the early period, 1976-1980, and the red dots pool all years in the late period, 2014-2018.

ventor. Recall that our measure of inventors is the number of successfully patenting inventors, which mechanically ensures that output per worker for single-inventor firms is at least one. Figure A3 corrects for this using employment histories of inventors from Revelio Labs to get a more accurate measure of inventors per firm (2008-2018), enabling us to include firm-year pairs with inventors but no patents, and shows that output per worker at single inventor firms is actually quite low. Otherwise the pattern is qualitatively similar to the USPTO figure.

Finally, the model predicts that larger firms will have a higher share of team patents in total patents. Defining firm size by the number of inventors, we estimate the following regression to examine the relationship between firm size and the propensity to engage in team patenting:

$$TeamPatShare_{ft} = \eta_f + \gamma_t + \log(Num.Inventors)_{ft} + \epsilon_{ft} \quad (A.1)$$

where $TeamPatShare_{ft}$ is the share of team patents out of total patents assigned by

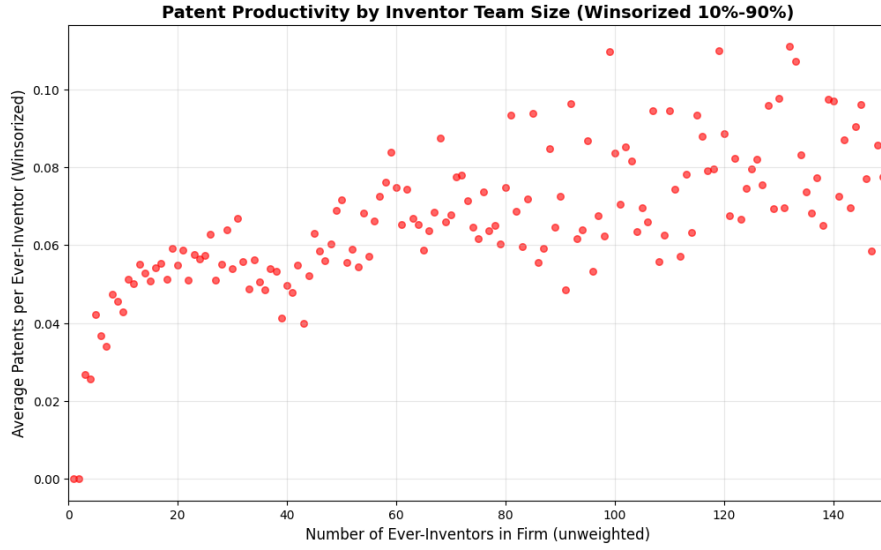


Figure A3: Number of patents per worker as a function of firm size in the Revelio Labs data, 2008-2018.

Note: The figure illustrates the number of patents per worker within firms, as a function of the total number of inventors at the firm in that year. The data is sourced from the Revelio Labs data, pooling all years between 2008 and 2018.

firm f in year t ; $\log(\text{Num.Inventors})_{ft}$ is the log number of inventors within firm f in t ; and η_f and γ_t are firm and year fixed effects, respectively.

Table A1 presents the results, where the first column uses all USPTO assignees, and the second column restricts the sample to public firms in Compustat. Across both samples, we find that larger firms exhibit a higher share of team patents than smaller firms. The effect is economically meaningful: a 1% increase in the number of inventors within a firm is associated with a 20.4 pp increase in the share of team patents for USPTO firms, and a 12 pp for public firms.

A.2 Teams of Three

In our baseline model each pair of inventors within a firm gets a bilateral draw and an inventor's productivity is the maximum over all their pairwise draws. With teams of three, every triple of inventors gets a draw, and an inventor's productivity is the maximum over all triples they belong to. With n workers, each worker belongs to

Table A1: Team Patent Share and Firm Size

	(1)	(2)
	All Firms	Public Firms
Log num. inventors	0.204*** (282.52)	0.122*** (22.46)
N of Obs.	717,111	74,501
Firm FE	Yes	Yes
Year FE	Yes	Yes
Adj. R^2	0.491	0.375

Note: The table presents the regression result from equation (A.1), where the dependent variable is the share of team patents within a firm. The coefficient of interest is on the log number of inventors in the firm. Firm and year fixed effects are included, respectively. Column (1) covers all USPTO assignees, and Column (2) includes Compustat firms only. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

$C(n-1, 2)$ distinct triples ("choose 2 out of $n-1$ coworkers"). $C(n-1, 2) = (n-1)(n-2)/2$, which grows like $\frac{n^2}{2}$ for large n . So instead of n^γ scaling, teams of three would deliver $(n^2)^\gamma = n^{2\gamma}$ scaling, making inventor productivity scale faster in the number of coworkers than in the baseline model.

A.3 Microfounded Team-Level Idea Production Function

This section shows how a version of the team production model of Freund (2024) generates stochastic idea quality from team production whose distribution resembles the Pareto distribution in equation 3.1.

The environment is the same as in Freund (2024), though we abstract from talent heterogeneity among inventors. Two inventors i and j produce an idea. Idea production requires a continuum (measure one) of tasks (for example researching the state of the art, defining the problem, sketching a solution). The quality of the idea is given by

$$z_{ij} = \left(\int_0^1 q(\tau)^{\frac{\eta-1}{\eta}} d\tau \right)^{\frac{\eta}{\eta-1}},$$

where $q(\tau)$ is the amount of task τ used and η is the elasticity of substitution across tasks. The amount of the task input $q(\tau)$ depends on the labor input $\ell_k(\tau)$ and task-level productivity $z_k(\tau)$ of each worker:

$$q(\tau) = z_i(\tau)l_i(\tau) + z_j(\tau)l_j(\tau).$$

Each worker has one unit of total time: $\int_0^1 \ell_k(\tau) d\tau = 1$.

Worker talent on each task $z_k(\tau)$ is drawn iid from a Frechet distribution:

$$P(z_k(\tau) < z) = \exp\left(-\left(\frac{z}{\iota x}\right)^{-\frac{1}{\chi}}\right),$$

where x is a common scale parameter that shifts inventor talent for all inventors. [Freund \(2024\)](#) instead allows overall talent x to vary across individuals but this is not needed for our results. χ governs specialization: higher χ indicates worker skills are more variable across tasks. $\iota = \Gamma(1 - \chi(1 - \eta))^{1/(1-\eta)}$ is a normalization that ensures changing χ does not mechanically change average productivity. Throughout we assume $1 + \chi(1 - \eta) > 0$.

[Freund \(2024\)](#) shows that the output of any pair of workers does not require information on task level productivities and instead is given by:

$$z_{ij} \sim 2^{1+\chi\xi_{ij}} x, \tag{A.2}$$

where ξ_{ij} is [Freund \(2024\)](#)'s measure of pairwise skill distance between the two workers, which is distributed uniformly $U[0, 1]$ in the population of inventors.

The key insight is the following: for a given skill distance ξ_{ij} , greater specialization of knowledge χ increases the quality of the idea produced by the team because they gain more from specializing across tasks. This effect is amplified if their skills are more distant from each other (ξ_{ij} is high). Talent x shifts the level of team output. [Figure A4](#) illustrates the distribution of realized idea quality using many draws (50,000) of pairwise skill distance and x normalized to 1. The dashed lines are the corresponding fitted Pareto distributions, where we estimate the Pareto tail parameter using the Hill maximum likelihood estimator on the realized idea quality data for each value of χ . Greater skill specialization χ implies a thicker right tail of the fitted Pareto, an increase in γ in our reduced form model.

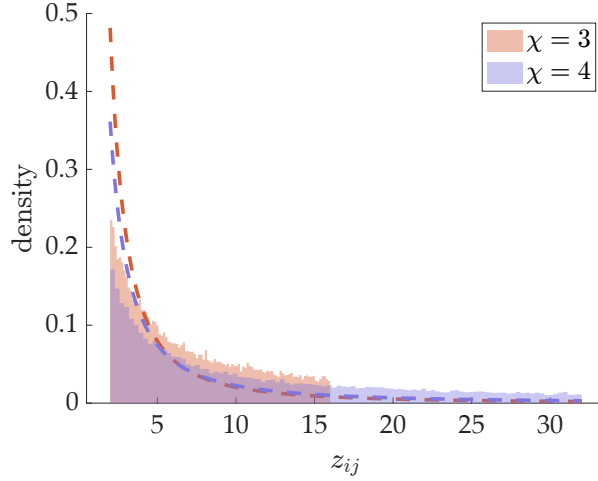


Figure A4: Distribution of idea quality z_{ij} (bars) and fitted Pareto (dashed lines) under two different values of specialization χ . Greater specialization implies more dispersion of output for the same draws of worker skill distance ξ_{ij} .

A.4 Derivation of Team Productivity

A.4.1 Discrete Firm Size

Recall that bilateral productivity is:

$$z_{ij} \sim \text{Pareto}(x, \gamma) : \quad P(z_{ij} \leq k) = 1 - \left(\frac{x}{k}\right)^\gamma \quad \forall k \geq x$$

Let the max of n draws be

$$M_n \equiv \max\{z_{1j}, z_{2j}, \dots, z_{nj}\}$$

then

$$F_{M_n}(k) \equiv P(M_n \leq k) = \left[1 - \left(\frac{x}{k}\right)^\gamma\right]^n \quad \forall k \geq x \quad (\text{A.3})$$

Then, we can derive the pdf as follows:

$$f_{M_n}(k) = n\gamma \left(\frac{x}{k}\right)^{\gamma+1} \left[1 - \left(\frac{x}{k}\right)^\gamma\right]^{n-1} \quad \forall k \geq x.$$

Thus, the expected value is

$$\mathbb{E}(M_n) = \int_k k f_{M_n}(k) dk = xnB(n, 1 - \gamma),$$

where B is Beta function.¹⁷

Note that for large n , the Beta function asymptotically becomes:

$$B(n, 1 - \gamma) = \frac{\Gamma(n)\Gamma(1 - \gamma)}{\Gamma(n + 1 - \gamma)} \simeq \Gamma(1 - \gamma)n^{-1+\gamma}.$$

Therefore, the expected productivity of working in a firm with n workers becomes:

$$\mathbb{E}(M_n) = x\Gamma(1 - \gamma)n^\gamma \tag{A.4}$$

which increases in n .

A.4.2 Continuous Firm Size

Assume that worker bilateral productivity is drawn from an inhomogeneous Poisson point process with intensity function $\lambda(k)$ within firm of size $n \in [0, \infty)$

$$\lambda(k) = n \frac{f(k)}{F(k)},$$

where $f(k)$ and $F(k)$ are the PDF and the CDF of the same Pareto distribution as in equation (3.1) so that the intensity function depends on the hazard rate of the Pareto distribution for team productivity. More workers (higher n) scales up the probability of productivity draws uniformly. The intuition of this process comes from rain falling on a sidewalk. If the rain is uniform across the sidewalk this is a homogeneous Poisson point process. The number of workers n scales up how heavy the rainfall is. Our choice of the particular intensity function over the real line (sidewalk) based on the Pareto

¹⁷It can be derived with a transformation of variable with $t = 1 - (x/k)^\gamma$. So that it can be rephrased as $\mathbb{E}(M_n) = nx \int_0^1 t^{n-1} (1 - t)^{-\gamma} dt$.

distribution in the model puts more “raindrops” on low productivity draws and ensures that even with a continuum of workers the maximum draw is finite.

Then the CDF of the maximum draw for worker i is given by the probability that no draw lies above k , $P[N(k, +\infty) = 0]$:

$$\begin{aligned}
F_{z_{it}^{team}}(k) &= \mathbb{P}[N(k, +\infty) = 0] \\
&= \exp \left(- \int_k^{+\infty} \lambda(u) du \right) \\
&= (\exp (-(\log F(\infty) - \log F(k))))^n \\
&= F(k)^n,
\end{aligned}$$

and thus, everything follows the same as in the discrete case from equation A.3 onward.

A.5 Proofs for Propositions

A.5.1 Proof for Proposition 1

We evaluate wedges in equation (4.6) at competitive equilibrium under solo production. Note that the firm decision rule in competitive equilibrium remains the same as in (3.9), except that the equilibrium wage is determined by the labor market clearing with solo production. The equilibrium wage is pinned down by

$$\int n^{solo}(\varepsilon_f) g(\varepsilon_f) = \bar{N},$$

which gives

$$w = \frac{1}{\bar{N}^{1-\alpha}} \alpha \zeta z^\alpha x_{\min}^f \left(\frac{\eta}{\eta - \frac{1}{1-\alpha}} \right)^{1-\alpha}.$$

We normalize $\bar{N} = 1$, and this gives a closed-form solution for firm labor demand

under solo production as follows:

$$n_f^{solo} = \frac{(\eta - \frac{1}{1-\alpha})\varepsilon_f^{\frac{1}{1-\alpha}}}{\eta(x_{\min}^f)^{\frac{1}{1-\alpha}}}.$$

This also determines spillovers and total output at the equilibrium,

$$\zeta = (x_{\min}^f)^\theta \left(\frac{\eta}{\eta - (\frac{1-\alpha+\beta}{1-\alpha})} \right)^\theta \left(\frac{\eta - \frac{1}{1-\alpha}}{\eta} \right)^{\beta\theta}$$

$$Y = \zeta z^\alpha x_{\min}^f \left(\frac{\eta}{\eta - \frac{1}{1-\alpha}} \right)^{1-\alpha}.$$

Plugging them into (4.6), we obtain the following expression for wedges under solo production:

$$\tau_f^{solo} = \frac{\theta\beta}{\alpha} (x_{\min}^f)^{\frac{\alpha-\beta}{1-\alpha}} \left(\frac{\eta - \frac{1-\alpha+\beta}{1-\alpha}}{\eta - \frac{1}{1-\alpha}} \right) \varepsilon_f^{-\frac{\alpha-\beta}{1-\alpha}}. \quad (\text{A.5})$$

Since $0 < \alpha < \beta < 1$, (A.5) is decreasing in firm productivity—and hence in employment size, given their positive relationship in competitive equilibrium.

A.5.2 Proof for Proposition 2

We evaluate wedges in equation (4.9) at competitive equilibrium under team production. As before, the equilibrium wage is pinned down by the following, which determines firm decision rule in (3.9) under team production:

$$\int n^{team}(\varepsilon_f) g(\varepsilon_f) = \bar{N},$$

which gives

$$w_t = \zeta \alpha (1 + \gamma) (x \Gamma (1 - \gamma))^\alpha x_{\min}^f \left(\frac{\eta}{\eta - \frac{1}{1-\alpha(1+\gamma)}} \right)^{\frac{1}{1-\alpha(1+\gamma)}}.$$

Thus, firm labor demand under team production is

$$n_f^{team} = \frac{(\eta - \frac{1}{1-\alpha(1+\gamma)}) \varepsilon_f^{\frac{1}{1-\alpha(1+\gamma)}}}{\eta (x_{\min}^f)^{\frac{1}{1-\alpha(1+\gamma)}}}.$$

This also determines spillovers and total output at the equilibrium,

$$\zeta = (x_{\min}^f)^\theta \left(\frac{\eta}{\eta - \frac{1-\alpha(1+\gamma)+\beta}{1-\alpha(1+\gamma)}} \right)^\theta \left(\frac{\eta - \frac{1}{1-\alpha(1+\gamma)}}{\eta} \right)^{\beta\theta}$$

$$Y = \zeta (x\Gamma(1-\gamma))^\alpha x_{\min}^f \left(\frac{\eta}{\eta - \frac{1}{1-\alpha(1+\gamma)}} \right)^{1-\alpha(1+\gamma)}.$$

Plugging them into (4.9), we obtain the following expression for wedges under team production:

$$\tau_f^{team} = \frac{\theta\beta}{\alpha(1+\gamma)} (x_{\min}^f)^{\frac{\alpha(1+\gamma)-\beta}{1-\alpha(1+\gamma)}} \left(\frac{\eta - \frac{1-\alpha(1+\gamma)+\beta}{1-\alpha(1+\gamma)}}{\eta - \frac{1}{1-\alpha(1+\gamma)}} \right) \varepsilon_f^{-\frac{\alpha(1+\gamma)-\beta}{1-\alpha(1+\gamma)}}. \quad (\text{A.6})$$

If $0 < \beta < \alpha(1+\gamma) < 1$, (A.6) is decreasing in firm productivity—and hence in employment size, given their positive relationship in competitive equilibrium.

A.5.3 Proof for Proposition 3

From equations (A.5) and (A.6), the wedges under solo and team production can be written in the unified form

$$\tau_f = C(\bar{\gamma}) \left(\frac{\varepsilon_f}{x_{\min}^f} \right)^{-\frac{\alpha(1+\bar{\gamma})-\beta}{1-\alpha(1+\bar{\gamma})}},$$

where $\bar{\gamma} = 0$ under solo production, $\bar{\gamma} = \gamma$ under team production, and

$$C(\bar{\gamma}) = \frac{\beta\theta}{\alpha(1+\bar{\gamma})} \left(\frac{\eta - \frac{1+\beta-\alpha(1+\bar{\gamma})}{1-\alpha(1+\bar{\gamma})}}{\eta - \frac{1}{1-\alpha(1+\bar{\gamma})}} \right)$$

is independent of firm productivity.

Taking logs gives

$$\log \tau_f = \log C(\bar{\gamma}) - \frac{\alpha(1 + \bar{\gamma}) - \beta}{1 - \alpha(1 + \bar{\gamma})} \log \left(\frac{\varepsilon_f}{x_{\min}^f} \right).$$

Hence, the elasticity of the wedge with respect to firm productivity is

$$\frac{\partial \log \tau_f}{\partial \log \varepsilon_f} = - \frac{\alpha(1 + \bar{\gamma}) - \beta}{1 - \alpha(1 + \bar{\gamma})}.$$

Differentiating with respect to $\bar{\gamma}$ yields

$$\frac{\partial}{\partial \bar{\gamma}} \left(\frac{\partial \log \tau_f}{\partial \log \varepsilon_f} \right) = - \frac{\alpha(1 - \beta)}{(1 - \alpha(1 + \bar{\gamma}))^2} < 0,$$

where the inequality follows from

$$0 < \beta < \alpha < \alpha(1 + \gamma) < 1.$$

Therefore, the elasticity of wedges with respect to firm productivity is more negative under team production than under solo production. Equivalently, wedges decline more rapidly with firm productivity under team production, implying a steeper relationship between wedges and firm productivity.

A.6 The Social Planner's Problem with Endogenous Firm Mass

This section presents the full-fledged setup of the social planner's problem with endogenous firm entry and exit. The following shows the planner's maximization:

$$\max_{M, A, n_f, \text{type}_f} M \int_A \zeta \varepsilon_f (z_f n_f)^\alpha df$$

subject to

$$M \left(\int_A (n_f + c) df + \kappa \right) = \bar{N}, \quad \zeta = \left(M \int_A \varepsilon_f n_f^\beta df \right)^\theta,$$

where M denotes the mass of firms, A denotes the set of active firms (equivalently, the choice of an exit cutoff), and type_f indicates whether firm f operates as a team or as a solo firm. The planner maximizes aggregate output, accounting for both the operating cost c and the entry cost κ , which are paid in units of labor.

First, the social planner chooses the mass of firms M and incurs a fixed entry cost κ per firm. After entry, each firm f draws an idiosyncratic productivity shock ε_f . Conditional on ε_f , the planner assigns to each firm f a productivity type—solo or team—with labor efficiency given by

$$\begin{aligned} z_f^{\text{solo}} &= z, \\ z_f^{\text{team}} &= x\Gamma(1 - \gamma)n_f^\gamma, \end{aligned}$$

and chooses its labor input n_f . The planner can instead shut down the firm. Each active firm $f \in A$ incurs a fixed cost c .

Let λ denote the Lagrange multiplier on the labor market clearing constraint. The Lagrangian associated with the social planner's problem is given by

$$\mathcal{L} = M^{1+\theta} \left(\int_A \varepsilon_f n_f^\beta df \right)^\theta \left(\int_A \varepsilon_f (z_f n_f)^\alpha df \right) - \lambda M \left(\int_A (n_f + c) df + \kappa \right) + \lambda \bar{N}.$$

For expositional simplicity, let

$$Z^{\text{raw}} = \int_A \varepsilon_f n_f^\beta df, \quad Y^{\text{raw}} = \int_A \varepsilon_f (z_f n_f)^\alpha df, \quad Y = M^{1+\theta} (Z^{\text{raw}})^\theta (Y^{\text{raw}}).$$

The first-order condition for a solo firm's labor choice is given by

$$\frac{Y}{M} \left(\theta \beta \frac{\varepsilon_f (n_f^{solo})^{\beta-1}}{Z^{raw}} + \alpha \frac{\varepsilon_f (z_f^{solo})^\alpha (n_f^{solo})^{\alpha-1}}{Y^{raw}} \right) = \lambda. \quad (\text{A.7})$$

The first-order condition for a team firm's labor choice is

$$\frac{Y}{M} \left(\theta \beta \frac{\varepsilon_f (n_f^{team})^{\beta-1}}{Z^{raw}} + \alpha (1 + \gamma) \frac{\varepsilon_f (z_f^{team})^\alpha (n_f^{team})^{\alpha-1}}{Y^{raw}} \right) = \lambda. \quad (\text{A.8})$$

The marginal benefits from assigning firms to each production type are given by the optimized surplus under each technology as follows:

$$\begin{cases} \frac{Y}{M} \left(\theta \frac{\varepsilon_f (n_f^{solo})^\beta}{Z^{raw}} + \frac{\varepsilon_f (z_f^{solo})^\alpha (n_f^{solo})^\alpha}{Y^{raw}} \right) - \lambda (n_f^{solo} + c), & \text{for solo firms,} \\ \frac{Y}{M} \left(\theta \frac{\varepsilon_f (n_f^{team})^\beta}{Z^{raw}} + \frac{\varepsilon_f (z_f^{team})^\alpha (n_f^{team})^\alpha}{Y^{raw}} \right) - \lambda (n_f^{team} + c), & \text{for team firms,} \\ 0, & \text{for inactive firms.} \end{cases} \quad (\text{A.9})$$

The social planner assigns firm f to the production type that yields the highest marginal benefit.

Finally, the first-order condition for choosing the number of firms is

$$(1 + \theta) \frac{Y}{M} - \lambda \left(\int_A (n_f + c) df + \kappa \right) = 0. \quad (\text{A.10})$$

The solution to the social planner's problem is jointly characterized by conditions (A.7)–(A.10) together with the labor market clearing condition.

The derivation of (A.9) is as follows. For simplicity, consider the following prototype problem:

$$J = \theta \log \int f(c, \varepsilon) d\varepsilon + \log \int g(c, \varepsilon) d\varepsilon, \quad c(\varepsilon) \in \{0, 1\}.$$

Consider a (measurable) perturbation $\delta c(\varepsilon) \in \{-1, 0, 1\}$ to the choice of firm types such

that $\int |\delta c(\varepsilon)| d\varepsilon$ is small. Let $\Delta f(\varepsilon)$ denote the difference $f(1, \varepsilon) - f(0, \varepsilon)$. Then,

$$\delta \int f(c, \varepsilon) d\varepsilon = \int \Delta f(\varepsilon) \delta c(\varepsilon) d\varepsilon.$$

This is also small when $\Delta f(\varepsilon)$ is bounded. Thus,

$$\delta \log \int f(c, \varepsilon) d\varepsilon \doteq \frac{\int \Delta f(\varepsilon) \delta c(\varepsilon) d\varepsilon}{\int f(c, \varepsilon) d\varepsilon}.$$

Similar results hold for g . Therefore, the first-order difference is

$$\begin{aligned} \delta J &\doteq \theta \frac{\int \Delta f(\varepsilon) \delta c(\varepsilon) d\varepsilon}{\int f(c, \varepsilon) d\varepsilon} + \frac{\int \Delta g(\varepsilon) \delta c(\varepsilon) d\varepsilon}{\int g(c, \varepsilon) d\varepsilon} \\ &= \int \left(\theta \frac{\Delta f(\varepsilon)}{\int f(c, \varepsilon) d\varepsilon} + \frac{\Delta g(\varepsilon)}{\int g(c, \varepsilon) d\varepsilon} \right) \delta c(\varepsilon) d\varepsilon. \end{aligned}$$

By the standard argument, this difference is necessarily zero for all valid $\delta c(\varepsilon)$ around the optimal choice of firm types. Finally, the first-order condition for optimality is

$$\theta \frac{\Delta f(\varepsilon)}{\int f(c, \varepsilon) d\varepsilon} + \frac{\Delta g(\varepsilon)}{\int g(c, \varepsilon) d\varepsilon} = 0.$$

Substituting into the expressions for f and g yields (A.9). Intuitively, the marginal contribution of changing a firm's type is given by a weighted sum of its effects on the spillover component and the productivity component.

Then, we can prove the following proposition, suggesting the existence of the two productivity cutoffs in the planner's equilibrium.

Proposition A4. *Suppose that $0 < \alpha, \beta < (1 + \gamma)^{-1}$. When inactive, solo, and team firms coexist, there exist cutoffs $\tilde{\varepsilon}_{spp}^{solo}$ and $\tilde{\varepsilon}_{spp}^{team}$ such that*

$$x < \tilde{\varepsilon}_{spp}^{solo} < \tilde{\varepsilon}_{spp}^{team}.$$

Firms are inactive if $\varepsilon_f < \tilde{\varepsilon}_{spp}^{solo}$, operate as solo firms if $\tilde{\varepsilon}_{spp}^{solo} \leq \varepsilon_f < \tilde{\varepsilon}_{spp}^{team}$, and operate as

team firms if $\varepsilon_f \geq \tilde{\varepsilon}_{spp}^{team}$. Moreover, at the solo–team cutoff $\tilde{\varepsilon}_{spp}^{team}$, we have

$$z_f^{solo} < z_f^{team} \quad \text{and} \quad n_f^{solo} < n_f^{team}.$$

Proof. Fix λ, Y, M, Z^{raw} , and Y^{raw} . Equations (A.7) and (A.8) imply that n^{solo} and n^{team} are continuously differentiable and strictly increasing in ε . The same holds for z^{solo} and z^{team} , and therefore for the marginal benefit functions MB^{solo} and MB^{team} in (A.9). Hence, at the solo–team cutoff $\tilde{\varepsilon}_{spp}^{team}$, the planner must be indifferent between assigning the firm to solo or team production, which requires $MB^{solo}(\tilde{\varepsilon}_{spp}^{team}) = MB^{team}(\tilde{\varepsilon}_{spp}^{team})$.

First, we show that $z_f^{solo} < z_f^{team}$ and $n_f^{solo} < n_f^{team}$ at the solo–team cutoff $\tilde{\varepsilon}_{spp}^{team}$. Let μ denote the common marginal benefit of assigning a firm to solo and team production at this cutoff. Then (z_f^{solo}, n_f^{solo}) and (z_f^{team}, n_f^{team}) solve, respectively, the same system of equations evaluated at $\eta = 0$ and $\eta = \gamma$:

$$\begin{aligned} \frac{Y}{M} \left(\theta \beta \frac{\tilde{\varepsilon}_{spp}^{team} n_f^{\beta-1}}{Z^{raw}} + \alpha(1+\eta) \frac{\tilde{\varepsilon}_{spp}^{team} z_f^\alpha n_f^{\alpha-1}}{Y^{raw}} \right) &= \lambda, \\ \frac{Y}{M} \left(\theta(1-\beta) \frac{\tilde{\varepsilon}_{spp}^{team} n_f^\beta}{Z^{raw}} + (1-\alpha(1+\eta)) \frac{\tilde{\varepsilon}_{spp}^{team} z_f^\alpha n_f^\alpha}{Y^{raw}} \right) &= \lambda c + \mu. \end{aligned}$$

The second equation is obtained by substituting (A.7) and (A.8) into the marginal benefit expressions in (A.9). Differentiating this system with respect to η yields

$$\begin{aligned} \beta \Psi_Z (\beta - 1) \frac{n'_f}{n_f} + \alpha_\eta \Psi_Y \left(\alpha \frac{z'_f}{z_f} + (\alpha - 1) \frac{n'_f}{n_f} \right) &= -\alpha \Psi_Y, \\ (1 - \beta) \Psi_Z \beta \frac{n'_f}{n_f} + (1 - \alpha_\eta) \Psi_Y \left(\alpha \frac{z'_f}{z_f} + \alpha \frac{n'_f}{n_f} \right) &= \alpha \Psi_Y, \end{aligned}$$

where,

$$\Psi_Z = \frac{Y}{M} \theta \frac{\tilde{\varepsilon}_{spp}^{team} n_f^\beta}{Z^{raw}}, \quad \Psi_Y = \frac{Y}{M} \frac{\tilde{\varepsilon}_{spp}^{team} z_f^\alpha n_f^\alpha}{Y^{raw}}, \quad \alpha_\eta = \alpha(1+\eta), \quad z'_f = \frac{dz_f}{d\eta}, \quad n'_f = \frac{dn_f}{d\eta}.$$

By Cramer's rule, we find that

$$\frac{z'_f}{z_f} = \frac{\eta}{D}(\alpha\Psi_Y)^2 > 0, \quad \frac{n'_f}{n_f} = \frac{1}{D}(\alpha\Psi_Y)^2 > 0,$$

where, the determinant is given by

$$D = \alpha\Psi_Y (\beta(1 - \beta)\Psi_Z + \alpha_\eta(1 - \alpha_\eta)\Psi_Y) > 0.$$

Therefore, $z_f^{solo} < z_f^{team}$ and $n_f^{solo} < n_f^{team}$.

Next, we show that the solo-versus-team cutoff $\tilde{\varepsilon}_{spp}^{team}$ is unique whenever it exists, i.e., MB^{solo} and MB^{team} can cross at most once. A sufficient condition is that whenever $MB^{solo}(\tilde{\varepsilon}_{spp}^{team}) = MB^{team}(\tilde{\varepsilon}_{spp}^{team})$, it holds that

$$\left. \frac{dMB^{solo}}{d\varepsilon} \right|_{\varepsilon=\tilde{\varepsilon}_{spp}^{team}} < \left. \frac{dMB^{team}}{d\varepsilon} \right|_{\varepsilon=\tilde{\varepsilon}_{spp}^{team}}.$$

We establish this by directly evaluating the derivatives at the cutoff $\tilde{\varepsilon}_{spp}^{team}$. The derivatives are, respectively,

$$\begin{aligned} \frac{dMB^{solo}}{d\varepsilon} &= \left(\frac{Y}{M} \left(\theta\beta \frac{\varepsilon(n_f^{solo})^{\beta-1}}{Z^{raw}} + \alpha \frac{\varepsilon(z_f^{solo})^\alpha (n_f^{solo})^{\alpha-1}}{Y^{raw}} \right) - \lambda \right) \frac{dn^{solo}}{d\varepsilon} \\ &\quad + \frac{Y}{M} \left(\theta \frac{(n_f^{solo})^\beta}{Z^{raw}} + \frac{(z_f^{solo})^\alpha (n_f^{solo})^\alpha}{Y^{raw}} \right) \\ &= \frac{Y}{M} \left(\theta \frac{(n_f^{solo})^\beta}{Z^{raw}} + \frac{(z_f^{solo})^\alpha (n_f^{solo})^\alpha}{Y^{raw}} \right), \\ \frac{dMB^{team}}{d\varepsilon} &= \left(\frac{Y}{M} \left(\theta\beta \frac{\varepsilon(n_f^{team})^{\beta-1}}{Z^{raw}} + \alpha \left(1 + \frac{1}{\gamma} \right) \frac{\varepsilon(z_f^{team})^\alpha (n_f^{team})^{\alpha-1}}{Y^{raw}} \right) - \lambda \right) \frac{dn^{team}}{d\varepsilon} \\ &\quad + \frac{Y}{M} \left(\theta \frac{(n_f^{team})^\beta}{Z^{raw}} + \frac{(z_f^{team})^\alpha (n_f^{team})^\alpha}{Y^{raw}} \right) \\ &= \frac{Y}{M} \left(\theta \frac{(n_f^{team})^\beta}{Z^{raw}} + \frac{(z_f^{team})^\alpha (n_f^{team})^\alpha}{Y^{raw}} \right). \end{aligned}$$

Here, we substitute (A.7) and (A.8) and cancel the first terms. At the cutoff $\tilde{\varepsilon}_{spp}^{team}$, the

marginal benefits coincide. Let μ denote the common value, i.e.,

$$\begin{aligned}\mu &= \frac{Y}{M} \left(\theta \frac{\tilde{\varepsilon}_{spp}^{team} (n_f^{solo})^\beta}{Z^{raw}} + \frac{\tilde{\varepsilon}_{spp}^{team} (z_f^{solo})^\alpha (n_f^{solo})^\alpha}{Y^{raw}} \right) - \lambda(n_f^{solo} + c), \\ &= \frac{Y}{M} \left(\theta \frac{\tilde{\varepsilon}_{spp}^{team} (n_f^{team})^\beta}{Z^{raw}} + \frac{\tilde{\varepsilon}_{spp}^{team} (z_f^{team})^\alpha (n_f^{team})^\alpha}{Y^{raw}} \right) - \lambda(n_f^{team} + c).\end{aligned}$$

Therefore, evaluating the derivatives at the cutoff $\tilde{\varepsilon}_{spp}^{team}$, we obtain the following:

$$\begin{aligned}\left. \frac{dMB^{solo}}{d\varepsilon} \right|_{\varepsilon=\tilde{\varepsilon}_{spp}^{team}} &= \frac{1}{\tilde{\varepsilon}_{spp}^{team}} (\mu + \lambda c + \lambda n_f^{solo}), \\ \left. \frac{dMB^{team}}{d\varepsilon} \right|_{\varepsilon=\tilde{\varepsilon}_{spp}^{team}} &= \frac{1}{\tilde{\varepsilon}_{spp}^{team}} (\mu + \lambda c + \lambda n_f^{team}).\end{aligned}$$

Since at this cutoff $z_f^{solo} < z_f^{team}$ and $n_f^{solo} < n_f^{team}$, it follows that

$$\left. \frac{dMB^{solo}}{d\varepsilon} \right|_{\varepsilon=\tilde{\varepsilon}_{spp}^{team}} < \left. \frac{dMB^{team}}{d\varepsilon} \right|_{\varepsilon=\tilde{\varepsilon}_{spp}^{team}}.$$

As previously shown, this implies that the marginal benefit functions MB^{solo} and MB^{team} can cross at most once. Therefore, the solo–team cutoff is unique whenever it exists.

Finally, we show that the exit cutoff $\tilde{\varepsilon}_{spp}^{solo}$ is unique whenever it exists. Substituting (A.7) and (A.8) into the marginal benefit expressions in (A.9) implies that

$$\begin{aligned}MB_f^{solo} &= \frac{Y}{M} \left(\theta(1-\beta) \frac{\varepsilon_f (n_f^{solo})^\beta}{Z^{raw}} + (1-\alpha) \frac{\varepsilon_f (z_f^{solo})^\alpha (n_f^{solo})^\alpha}{Y^{raw}} \right) - \lambda c, \\ MB_f^{team} &= \frac{Y}{M} \left(\theta(1-\beta) \frac{\varepsilon_f (n_f^{team})^\beta}{Z^{raw}} + \theta(1-\alpha(1+\gamma)) \frac{\varepsilon_f (z_f^{team})^\alpha (n_f^{team})^\alpha}{Y^{raw}} \right) - \lambda c.\end{aligned}$$

Regarding both as a function of ε_f , each is strictly increasing in ε_f . Hence, their point-wise maximum can cross the horizontal line $MB = 0$ at most once. Therefore, the exit cutoff is unique whenever it exists. $\square$

Table B2: Model Predicted Changes in Aggregate Moments, $\eta = 0.145$

	Early	Late	Late + η
Inventor wage w	0.724	0.894	0.724
Prod. cutoff (exit)	1.004	1.263	1.023
Prod. cutoff (team)	1.580	1.657	1.316
Share of solo-based firms	93.7	80.9	82.3
Share of team patents	57.8	92.7	77.6
Employment share of firms with $n > 145$	32.8	79.6	44.0
Average firm size	7.1	30.2	9.5
Spillovers $z^{spillover}$	0.903	0.823	0.887
Output (normalized)	1.000	1.234	1.000

Notes: This table reports various aggregate moments of the competitive equilibrium. The first column is the early period calibration. The second column is the baseline late period calibration. The third column uses $\eta = 0.145$ in the late period (targeting employment concentration) and changes the idea production parameters as in Table ehtab:param.

B Supplementary Model Figures and Tables

B.1 Experiments with firm productivity distribution (η)

This Appendix presents results from two additional experiments where η , the shape of the Pareto distribution of firm productivities ε_f is allowed to vary between the early and late period. In the first experiment, we vary the idea production parameters as in Table 3 and then choose a new η to exactly fit the level of employment concentration in the late period. This involves choose a thinner tail for the firm productivity distribution with lower η , lowering mean productivity of firms in the economy, because the idea production function changes overstate the rise in employment concentration, and less dispersed firm productivities in the late period are needed to lower employment concentration back down to the value observed in the data.

The results are in Table B2. Lower average firm productivity reduces inventors wages, contrary to the data, and output falls. The joint changes, though, get closer to matching the increase in firm size in the data (10.3 in the data), and the team patent share (77.6% in the data).

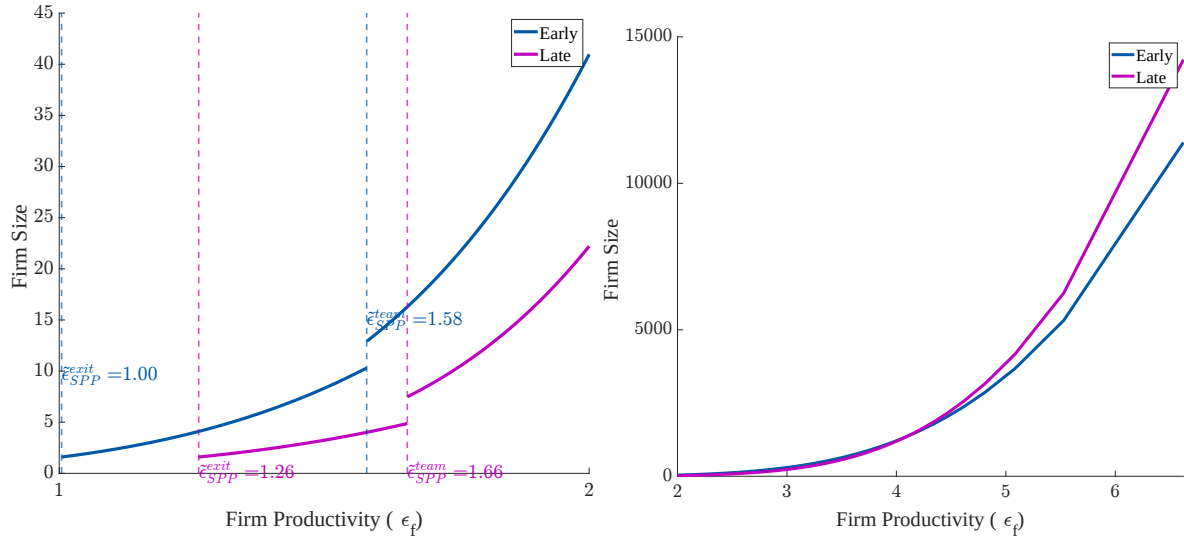
Table B3: Model Predicted Changes in Aggregate Moments, $\eta = 0.173$

	Early	Late	$\eta = 0.173$
Inventor wage w	0.724	0.894	0.755
Prod. cutoff (exit)	1.004	1.263	1.058
Prod. cutoff (team)	1.580	1.657	1.665
Share of solo-based firms	93.7	80.9	92.7
Share of team patents	57.8	92.7	66.7
Employment share of firms with $n > 145$	32.8	79.6	44.0
Average firm size	7.1	30.2	9.1
Spillovers $z^{spillover}$	0.903	0.823	0.894
Output (normalized)	1.000	1.234	1.043

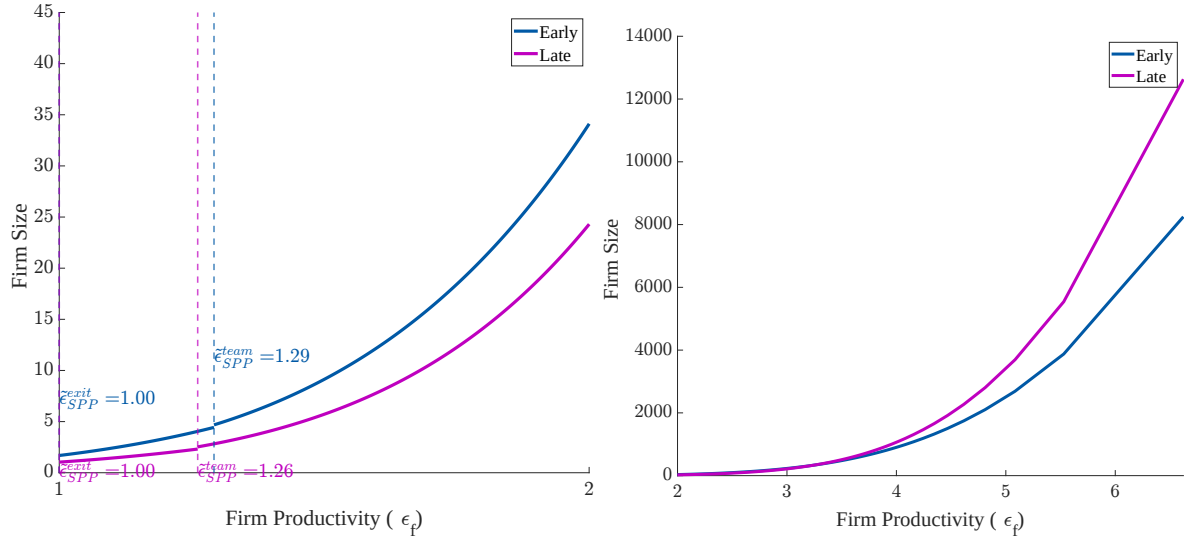
Notes: This table reports various aggregate moments of the competitive equilibrium. The first column is the early period calibration. The second column is the baseline late period calibration. The third column uses $\eta = 0.173$ in the late period (targeting employment concentration) and holds the idea production parameters fixed to their early period values in Table eatab:param.

In a second experiment, we change η by itself holding the idea production function parameters fixed at their early values to target employment concentration in the late period. In this case a higher value of $\eta = 0.173$ is need to match the rise in concentration by making the firm productivity distribution more skewed (and raising average firm productivity). The results are in Table B3. The change in the firm productivity distribution to match higher concentration is consistent with a rise in the team patenting share and a modest rise in firm size, consistent with the data. However, the share of solo-based firms is almost unaffected, unlike in the data.

B.2 Size Decision Rules



(a) Competitive Equilibrium



(b) Social Optimum

Figure B5: Size Decision Rule: CE (top) vs. SPP (bottom) in Early and Late Periods

Note: These figures illustrate firm-size decision rules under the competitive equilibrium (top panel) and the social planner's allocation (bottom panel), for both the early and late periods. In each panel, the left subfigure focuses on the low-productivity (small-firm) region, while the right subfigure focuses on the high-productivity (large-firm) region to highlight the underlying patterns clearly. The blue lines correspond to the early period, and the purple lines indicate the late period. The vertical lines denote two productivity cutoffs for adopting team production and firm exit in each equilibrium for each period.

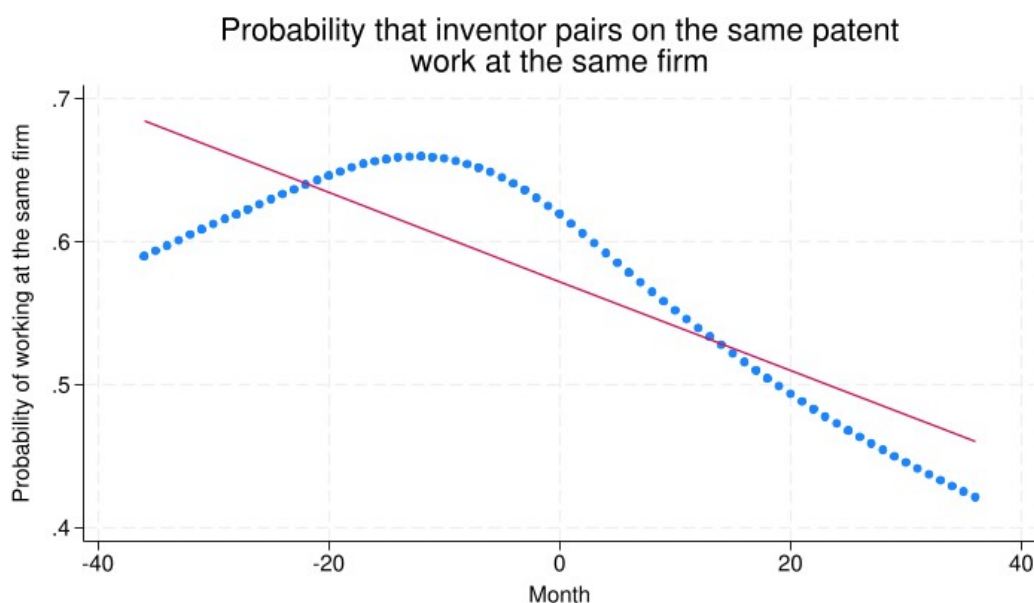


Figure C6: The probability for a pair of inventors on a patent working at the same firm 36 months before and after the filing month. $t = 0$ corresponds to the filing month of the patent.

C Empirical Appendix

C.1 Probability of Inventors Working at the Same Firm

We use patent data to measure how often inventors listed on the same team patent are employed by the same firm around the time of invention. Specifically, we identify all pairs of inventors listed on each team patent and construct a monthly panel spanning five years before and after the patent filing date. For each inventor pair and month, we define an indicator equal to one if both inventors are employed by the same firm in that month and zero otherwise.

Based on this indicator, we compute the average probability that a pair of inventors is employed by the same firm over a five-year window before and after the patent filing date. We estimate this probability to be 78.75%.

Furthermore, in Figure C6, we plot the average value of this indicator by month relative to the patent filing date, which measures the probability that a given pair of

inventors collaborating on a patent is employed by the same firm as a function of time relative to the filing date. The probability that the two inventors are employed by the same firm peaks at approximately 0.66 twelve months before the patent filing date and declines steadily thereafter, reaching approximately 0.40 three years after the filing date. This pattern suggests that inventor teams are more likely to be formed within the same firm as the invention approaches and frequently separate following the completion of a joint invention.

C.2 Robustness for Idea Production Function Estimation

Table C4: Returns to Scale in Team Production: Citations

	(1)	(2)
	All	Public Firms
$\log(n_{ft})$	0.575*** (0.003)	0.686*** (0.006)
$\log(n_{ft}) \times \text{Team}_{ft}$	0.001 (0.004)	0.041*** (0.007)
$\log(n_{ft}) \times \text{Team}_{ft} \times \text{Post}_t$	0.022*** (0.005)	0.003 (0.008)
Team_{ft}	-0.429*** (0.005)	-0.472*** (0.011)
Post_t	-0.095*** (0.003)	-0.105*** (0.012)
$\log(n_{ft}) \times \text{Post}_t$	0.009*** (0.003)	0.028*** (0.005)
$\text{Team}_{ft} \times \text{Post}_t$	-0.042*** (0.006)	-0.014 (0.015)
$\log(K_{ft}^{\text{intan.}})$		-0.040*** (0.003)
Constant	0.613*** (0.003)	0.774*** (0.021)
R-squared	0.452	0.616
Obs.	1105437	131166
Firm FE	Yes	Yes

Notes: Column (1) includes all firm-year observations. Column (2) includes public firms with available intangible capital data from [Peters and Taylor \(2017\)](#). Outcome is 5-year forward citations purged of technology-class fixed effects. See equation (5.1) for details. Standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C5: Returns to Scale in Team Production: Revelio Labs Data

	(1)	(2)
	All	$n > 1$
$\log(n_{ft})$	0.353*** (0.010)	0.457*** (0.012)
$\log(n_{ft}) \times \text{Team}_{ft}$	0.169*** (0.007)	0.161*** (0.009)
$\log(n_{ft}) \times \text{Team}_{ft} \times \text{Post}_t$	0.041*** (0.010)	0.055*** (0.014)
Team_{ft}	-0.071*** (0.014)	-0.096*** (0.019)
Post_t	0.006 (0.013)	0.003 (0.025)
$\log(n_{ft}) \times \text{Post}_t$	0.019** (0.009)	0.009 (0.014)
$\text{Team}_{ft} \times \text{Post}_t$	0.023 (0.018)	-0.009 (0.027)
Constant	0.026* (0.016)	-0.254*** (0.025)
R-squared	0.244	0.275
Obs.	169,800	135,406
Firm FE	Yes	Yes

Notes: Column (1) includes all firm-year observations. Column (2) restricts the sample to firm-year observations with at least two inventors. Outcome is number of patents. Data is Revelio Labs, 2008-2018, post-period is 2014-2018. See equation (5.1) for details. Standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.3 Model Validation

We examine the relationship between inventor concentration and team innovation using patent data. We construct these measures at the CPC technology group level and estimate the following regression:

$$\text{Concentration}_{ct} = \beta \text{TeamPatentShare}_{ct} + X'_{ct} \gamma + \mu_c + \delta_t + \epsilon_{ct}, \quad (\text{C.1})$$

where $\text{Concentration}_{ct}$ denotes the share of inventors at firms with more than 145 inventors in CPC technology group c and year t , $\text{TeamPatentShare}_{ct}$ is the average firm-

Table C6: Inventor Concentration and Team Patenting

	(1)	(2)	(3)	(4)	(5)	(6)
Team patent share	0.308*** (0.007)	0.210*** (0.007)	0.257*** (0.021)	0.186*** (0.018)	0.058*** (0.011)	0.053*** (0.010)
log(Firms)		-0.245*** (0.004)		-0.244*** (0.015)		-0.150*** (0.013)
log(Patents)		0.252*** (0.003)		0.251*** (0.013)		0.146*** (0.011)
R-squared	0.079	0.335	0.091	0.339	0.116	0.164
Obs.	25,961	25,961	25,961	25,961	25,961	25,961
Year FE	No	No	Yes	Yes	Yes	Yes
CPC FE	No	No	No	No	Yes	Yes

Notes: The dependent variable is inventor concentration at the CPC technology-group level. Team patent share is the fraction of patents within a CPC technology group that are produced by teams of inventors. Controls include the log number of firms and the log number of patents in the technology class. Standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

level share of team patents, and X_{ct} is a vector of controls, including the log number of firms and the log number of patents, all measured at the CPC technology group–year level. μ_c and δ_t denote CPC technology-group and year fixed effects, respectively. The coefficient of interest, β , measures the association between inventor concentration and the prevalence of team-based innovation within a technology class.

Table C6 reports the estimation results. We find a positive and statistically significant association between inventor concentration and the share of team patents across all specifications. The estimated relationship is robust to the inclusion of controls as well as year and CPC technology-class fixed effects and is consistent with our model predictions.

Furthermore, Figure C7 provides complementary evidence by examining changes in inventor concentration and team patenting across CPC technology groups over time. For each CPC technology group, we compute the average inventor concentration and the average patent-weighted firm-level team patent share separately for the periods 1976–1999 and 2000–2018. We then measure the change in each variable as the difference between its average in the later period and its average in the earlier period. The figure plots the change in inventor concentration against the corresponding change in

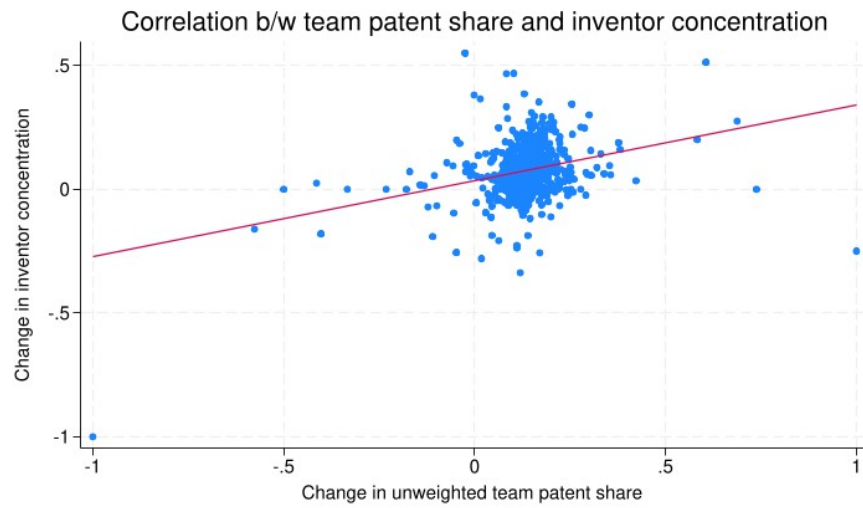


Figure C7: Inventor Concentration and Team Patent Share

the team patent share across CPC technology groups. The figure shows a positive relationship, indicating that technology groups experiencing larger increases in inventor concentration also tend to exhibit larger increases in the prevalence of team-based innovation.

C.4 Team Diversity

Another trend over the past decades is rising knowledge diversity within teams. To measure knowledge diversity, we first identify the “core” CPC subgroup associated with an inventor in a given year as a proxy for the inventor’s type or area of expertise.^{18,19} The core CPC subgroup of an inventor is defined as the subgroup with the highest number of patents filed by that inventor up to that year.²⁰ For each team, we calculate the number of core CPC subgroups associated with its inventors, which serves as a measure of the team’s knowledge diversity in that year. We then average this knowledge diversity measure across all teams within a firm for that year. Finally, we compute the average of these firm-level measures across all firms in a given year.

The result is plotted in Figure C8, which shows an upward trend and an increasing likelihood of the average team working with inventors with different specialties.

¹⁸For example, G06N 20/00 is “Machine learning” or A23B 11/00 is “Preservation of milk or dairy products.”

¹⁹For a patent associated with multiple CPC subgroups, we pick the main CPC subgroups based on the sequence variable provided by the USPTO.

²⁰Eventually, we aim to leverage the Longitudinal Employer-Household Dynamics (LEHD) dataset, which contains detailed demographic and educational information for workers covered by the U.S. Unemployment Insurance (UI) system. By linking it to the Longitudinal Business Database (LBD) and USPTO inventor data, we plan to track workers’ patenting activities within and across firms. This is in progress.

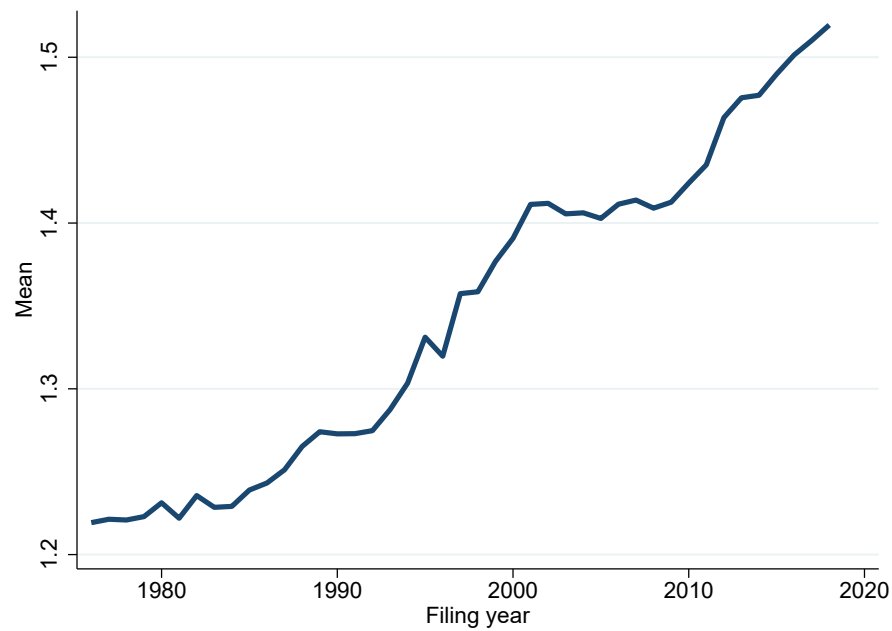


Figure C8: Average Number of Inventor CPC Subgroups in Teams in a Firm

Note: The figure illustrates the average number of core CPC subgroups associated with inventors on team patents in USPTO. A core CPC subgroup is defined as the modal CPC subgroup in an inventor's patenting history up to a given year. For each firm, we calculate the average number of core CPC subgroups for inventors on team patents and then compute the mean across all firms in a given year.